

Gradient descent for unbounded convex functions on Hadamard manifolds and its applications to scaling problems*

Hiroshi Hirai[†] Keiya Sakabe[‡]

January 13, 2025

Abstract

In this paper, we study asymptotic behaviors of continuous-time and discrete-time gradient flows of a “lower-unbounded” convex function f on a Hadamard manifold M , particularly, their convergence properties to the boundary M^∞ at infinity of M . We establish a duality theorem that the infimum of the gradient-norm $\|\nabla f(x)\|$ of f over M is equal to the supremum of the negative of the recession function f^∞ of f over the boundary M^∞ , provided the infimum is positive. Further, the infimum and the supremum are obtained by the limits of the gradient flows of f . Our results feature convex-optimization ingredients of the moment-weight inequality for reductive group actions by Georgoulas, Robbin, and Salamon, and are applied to noncommutative optimization by Bürgisser et al. FOCS 2019. We show that the gradient descent of the Kempf-Ness function for an unstable orbit converges to a 1-parameter subgroup in the Hilbert-Mumford criterion, and the associated moment-map sequence converges to the minimum-norm point of the moment polytope. We show further refinements for operator scaling—the left-right action on a matrix tuple $A = (A_1, A_2, \dots, A_N)$. We characterize the gradient-flow limit of operator scaling by a vector-space generalization of the classical Dulmage-Mendelsohn decomposition of a bipartite graph. Also, for a special case of $N = 2$, we reveal that this limit determines the Kronecker canonical form of matrix pencils $sA_1 + A_2$.

Keywords: Hadamard manifold, geodesically-convex optimization, gradient flow, matrix scaling, geometric programming, Hilbert-Mumford criterion, Kempf-Ness theorem, moment map, moment polytope, operator scaling, Dulmage-Mendelsohn decomposition, Kronecker canonical form, matrix pencil.

MSC-classifications: 90C25, 53C35

*The conference version is appeared as H. Hirai and K. Sakabe: Gradient descent for unbounded convex functions on Hadamard manifolds and its applications to scaling problems. *65th IEEE Symposium on Foundations of Computer Science (FOCS 2024)*, pp.2387–2402

[†]Graduate School of Mathematics, Nagoya University, Nagoya, 464-8602, Japan. hirai.hiroshi@math.nagoya-u.ac.jp

[‡]Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, 113-8656, Japan. ksakabe@g.ecc.u-tokyo.ac.jp

1 Introduction

In convex optimization, it is usually assumed that the objective function f is bounded below. The performance of a minimization algorithm is evaluated by convergence behavior to the minimum of f . The present paper addresses convergence behavior of a minimization algorithm for a “lower-unbounded” convex function f , i.e., $\inf f(x) = -\infty$. This may look meaningless, because the trajectory x_i of the algorithm diverges to the infinity, and $f(x_i)$ goes to $-\infty$. The meta question of the paper is:

What can we gain from such a divergent sequence?

Let us formalize our setting and mention its background. Let M be a Hadamard manifold—a simply-connected complete Riemannian manifold with nonpositive sectional curvature. Let $f : M \rightarrow \mathbb{R}$ be a differentiable (geodesically-)convex function, that is, f is convex along any geodesic. We consider the following unconstrained convex optimization on M :

$$\inf. \quad f(x) \quad \text{s.t.} \quad x \in M, \quad \text{where } f \text{ can be lower-unbounded.} \quad (1.1)$$

Such a problem setting is significant in the recent progress on *operator scaling* [26] and generalizations; see [1, 11, 12, 21, 22, 23, 30]. In the classical *matrix scaling* [48], the scalability is equivalent to the boundedness of (1.1) for some convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Further, it is also equivalent to the perfect matching condition of the associated bipartite graph. Hayashi, Hirai, and Sakabe [28] studies the asymptotic behavior of the Sinkhorn algorithm for unscalable (unbounded) case, and revealed that a combinatorial certificate (*Hall blocker*) of unscalability can be identified from the divergent behavior of the Sinkhorn algorithm. Although a Hall blocker is easily obtained by a network-flow algorithm, finding the corresponding certificate (*shrunk subspace*) in the operator scaling setting is possible but quite difficult; see [27, 33, 34]. A recent work by Franks, Soma, and Goemans [19] modifies the *operator Sinkhorn algorithm*—alternating minimization algorithm for a convex function on a Hadamard manifold of positive definite matrices—to obtain a shrunk subspace in polynomial time, although it is still rather complicated. The matrix and operator scaling problems are generalized to a class of convex optimization involving reductive group actions, called *noncommutative optimization* [12], which asks to minimize the *Kempf-Ness function* associated with an orbit of the action. This is formulated as convex optimization on a representative example of Hadamard manifolds—*symmetric space of nonpositive curvature*. It is lower-unbounded if and only if the orbit is unstable, where a 1-parameter subgroup (*destabilizing 1-PSG*) in the *Hilbert-Mumford criterion* is an unbounded certificate generalizing a Hall blocker and shrunk subspace. As mentioned in [12], the design of polynomial-time algorithms for several noncommutative optimizations, such as (un)stability determination, moment-polytope membership, and orbit-closure intersection, is a great challenge, and will bring fruitful applications to broader areas of mathematical sciences. Many of them involve the (un)bounded determination of Kempf-Ness functions, though our current knowledge on such problems is only limited.

Motivated by these considerations, we study the minimization of lower-unbounded convex functions on Hadamard manifolds. Even in the Euclidean setting $M = \mathbb{R}^n$, there are few works (see e.g., [4, 43]) on such study. We focus on asymptotic behavior of the most simplest algorithm—*gradient descent*. Accompanied with this, we also consider its continuous version—*gradient flow*, that is, a trajectory produced by differential equation $\dot{x}(t) = -\nabla f(x(t))$.

The contribution and organization of this paper are summarized as follows. We begin with a general study on asymptotic behavior of the gradient flow/descent for an unbounded convex function f on Hadamard manifold M . As in the Euclidean setting, the *recession function* (*asymptotic slope*) f^∞ of f (see [29, 36]) is a basic tool analyzing the unboundedness, which is a function defined on the *boundary* M^∞ *at infinity* of M . Intuitively, the boundary M^∞ is the set of all *directions* ξ from an arbitrary fixed point x_0 , and $f^\infty(\xi)$ represents the slope of f along the direction ξ at infinity. A Hadamard manifold M admits a compactification $M \cup M^\infty$, where the resulting topology is called the *cone topology*. These notions and related manifold terminologies are summarized in Section 2.

We focus on the convergence properties, with respect to the cone topology, of the gradient flow/descent for an unbounded convex function f . In Section 3, under the sufficient condition $\inf_{x \in M} \|\nabla f(x)\| > 0$ of unboundedness, we establish in Theorem 3.1 that the gradient flow $x(t)$ converges to a point of boundary M^∞ with providing the following min-max (inf-sup) relation:

$$\lim_{t \rightarrow \infty} \|\nabla f(x(t))\| = \inf_{x \in M} \|\nabla f(x)\| = \sup_{\xi \in M^\infty} -f^\infty(\xi) = -f^\infty(\lim_{t \rightarrow \infty} x(t)). \quad (1.2)$$

The limit $\lim_{t \rightarrow \infty} x(t)$ is the unique minimizer of f^∞ over M^∞ , and is a certificate of the unboundedness. Further, we also show in Theorem 3.7 that the same result holds for the sequence x_i obtained by the gradient descent applied to an L -smooth convex function f with step-size $1/L$. These are the core results of the paper that drives the subsequent arguments.

Even Euclidean setting $M = \mathbb{R}^n$, the convergence results on the gradient flow/descent seem new, and brings an interesting ramification (Theorem 3.14): both $\nabla f(x(t))$ and $\nabla f(x_i)$ converge to the minimum-norm point p^* of the gradient space $\nabla f(\mathbb{R}^n)$ (that is convex). This means that the gradient descent is interpreted as a minimum-norm point algorithm in the gradient space. Other interesting connections and implications to *Hessian Riemannian gradient flow* [2], *mirror descent* [42], and geometric programming are also mentioned.

In Section 4, we mention applications. In Section 4.1, we deal with the norm-minimization in a reductive group action on a complex vector/projective space. As mentioned, this is the minimization of the Kempf-Ness function F_v of an orbit. Then, the gradient descent in this case is nothing but the *first-order algorithm* in [12]. Applying our results, we show that for the unstable case the trajectory of the first-order algorithm converges, in cone topology, to the unique minimizer of F_v^∞ , that produces a destabilizing 1-PSG in the Hilbert-Mumford criterion. Further, the spectre of the moment-map (= transported gradient of F_v) along the trajectory converges to the minimum-norm point of the moment polytope Δ_v . For the gradient-flow setting, we reveal the connection to the theory of the *moment-weight inequality* for reductive group actions, developed by Georgoulas, Robbin, and Salamon [24] building upon the earlier work of GIT by Kempf, Kirwan, Mumford, Ness and a recent work by Chen and Sun [15, Section 4] on K -stability. Specifically, the weak duality $\|\nabla f(x)\| \geq -f^\infty(\xi)$ in (1.2) becomes the moment-weight inequality, and the strong duality via the gradient flow can explain important parts of their theory. It may be fair to say that our results in Section 3 extract and discretize convex-optimization ingredients of their theory.

In Section 4.2, we focus on the *left-right action* $SL_n(\mathbb{C}) \times SL_m(\mathbb{C}) \ni (g, h) \mapsto gAh^\dagger$ on a matrix tuple $A = (A_1, A_2, \dots, A_N)$, that corresponds to the operator scaling problem.

In this setting, the middle equality in (1.2) is interpreted as a duality theorem for the scalability limitation (Theorem 4.19), which sharpens Gurvits' characterization in an inf-sup form. We then study the limit of the gradient flow/descent of the Kempf-Ness function $(g, h) \mapsto \log \|gAh^\dagger\|$. Our focus is in the unscalable case, whereas the scalable case was studied in detail by Kwok, Lau, and Ramachandran [39]. We show in Theorems 4.22 and 4.24 that the minimum-norm point of the moment polytope Δ_A and the limit of the gradient flow/descent are characterized by a certain simultaneous block-triangularization of $A = (A_1, A_2, \dots, A_N)$, which is a vector-space generalization of the classical *Dulmage-Mendelsohn decomposition* [17] of a bipartite graph. More specifically, the sequence of (normalized) scaling tuples $g_t Ah_t^\dagger / \|g_t Ah_t^\dagger\|$ along the gradient flow/descent converges to a block-diagonal matrix modulo the left-right unitary group action, where the block structure is determined by our generalized DM-decomposition. This answers the gradient-flow/descent variant of an open question by Garg and Oliveria [23, Section 6] for asking the behavior of the operator Sinkhorn algorithm for unscalable instances. Finding this block-structure itself is significant. We partially eliminate the unitary indeterminacy from $g_t Ah_t^\dagger / \|g_t Ah_t^\dagger\|$, and exploit a convergence sequence to a coarse block-triangular structure (Theorems 4.25 and 4.26). This leads to a construction of a shrunk subspace (a certificate of unscalability) by the simple gradient descent and the rounding procedure of Franks, Soma, and Goemans [19].

In Section 4.3, for a special case of $N = 2$, we reveal that our DM-decomposition of (A_1, A_2) coarsens and determines the well-known *Kronecker canonical form* of matrix pencils $sA_1 + A_2$. The Kronecker form plays important roles in the systems analysis by a differential-algebraic equation (DAE) $A_1 \dot{u}(t) + A_2 u(t) = 0$. Its computation has been studied for a long time in the literature of numerical computation; see e.g., [16, 49]. Our convergence result (Theorem 4.30) suggests a new iterative method for determining the Kronecker structure, which is based on the simple gradient descent and is conceptually different from the existing ones.

These results may be positioned as attempts of detecting, by algorithms in M , hidden structures in the boundary M^∞ at infinity, which has been little studied so far. We hope that our attempts lead to more serious studies from computational complexity perspective. Particularly, it is an important future direction to improve the present convergence-type results of the gradient descent to the ones having explicit iteration complexity.

2 Preliminaries

Let $\mathbb{R}$ and $\mathbb{R}_+$ denote the sets of real and nonnegative real numbers, respectively. We often add to $\mathbb{R}$ and $\mathbb{R}_+$ the infinity elements $\pm\infty$, where the topology and ordering $\leq$ are extended in the natural way. Let $\mathbb{C}$ denote the set of complex numbers $z = x + iy$, where $\bar{z}$ denotes the complex conjugate $x - iy$ and $\operatorname{Re} z$ denotes the real part x . The same notation is used for a complex vector $\zeta = u + iv \in \mathbb{C}^n$ with $u, v \in \mathbb{R}^n$ as $\bar{\zeta} = u - iv$. For a matrix A over $\mathbb{C}$, let $A^\dagger$ denote the transpose conjugate. For sets I and J of row indices and column indices, let $A[I, J]$ denote the submatrix of A with row indices in I and column indices in J . For two matrices A, B (of possibly different sizes), let $A \oplus B$ denote the block-diagonal matrix of block-diagonals A, B in order. For a vector $p \in \mathbb{R}^n$, let $\operatorname{diag} p$ denote the $n \times n$ diagonal matrix with $(\operatorname{diag} p)_{ii} = p_i$.

The general linear group $GL(n, \mathbb{C})$ and special linear group $SL(n, \mathbb{C})$ over $\mathbb{C}$ are simply

denoted by GL_n and SL_n , respectively. The unitary group $U(n)$ and special unitary group $SU(n)$ are denoted by U_n and SU_n , respectively. For a finite dimensional vector space V over $\mathbb{C}$, let $GL(V)$ denote the group of linear isomorphisms on V .

For a positive integer n , let $[n] := \{1, 2, \dots, n\}$. For $X \subseteq [n]$, let $\mathbf{1}_X \in \mathbb{R}^n$ be defined as $(\mathbf{1}_X)_i = 1$ if $i \in X$ and 0 otherwise. If $X = [n]$, then $\mathbf{1}_{[n]}$ is simply written as $\mathbf{1}$.

Sequences $(x_i)_{i=0,1,\dots}$, and $(x(t))_{t \in \mathbb{R}_+}$ are simply denoted by x_i and $x(t)$, respectively.

2.1 Riemannian geometry

We will utilize standard terminologies and notation on Riemannian geometry; see e.g., [46]. See also a recent book [8] for optimization perspective. We assume sufficient differentiability for manifolds, and functions, maps, and vector/tensor fields on them. Let M be a Riemannian manifold. For $x \in M$, let $T_x = T_x(M)$ denote the tangent space of M at x , where $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_x$ denotes the Riemannian metric at x and $\|\cdot\| := \sqrt{\langle \cdot, \cdot \rangle}$ denotes the associated norm. Let $S_x := \{u \in T_x \mid \|u\| = 1\}$ and $B_x := \{u \in T_x \mid \|u\| \leq 1\}$ denote the unit sphere and ball in T_x , respectively. The angle $\angle(u, v)$ of two vectors $u, v \in T_x$ is defined as $\cos^{-1}(\langle u, v \rangle / \|u\| \|v\|)$. The product $M \times M'$ of two Riemannian manifolds M, M' is viewed as a Riemannian manifold by setting $\langle (u, u'), (v, v') \rangle_{(x, x')} := \langle u, v \rangle_x + \langle u', v' \rangle_{x'}$.

For a path $\gamma : [a, b] \rightarrow M$ and $t \in [a, b]$, let $\dot{\gamma}(t)$ denote the tangent vector of γ at $T_{\gamma(t)}$. The length of the path γ is defined by $\int_a^b \|\dot{\gamma}(t)\| dt$. The distance $d(x, y)$ between $x, y \in M$ is the infimum of the length of a path connecting x and y . We consider the Levi-Civita connection ∇ associated with the Riemannian metric. The connection ∇ determines a *parallel transport* $\tau_\gamma^t : T_{\gamma(0)} \rightarrow T_{\gamma(t)}$ along any path $\gamma : [0, b] \rightarrow M$ with $t \in [0, b]$, where $\tau_\gamma^{-t} := (\tau_\gamma^t)^{-1}$. By using the parallel transport, the covariant derivative $\nabla_u V$ of a vector field $V = (V_x)_{x \in M}$ by $u \in T_x$ is given by $\nabla_u V := (d/dt) \tau_\gamma^{-t} V_{\gamma(t)}|_{t=0}$, where γ is a path with $\gamma(0) = x$ and $\dot{\gamma}(0) = u$.

In this paper, any manifold M is assumed complete. That is, the metric space (M, d) are complete. Then the distance $d(x, y)$ is always attained by a *geodesic*—a path $\gamma : [a, b] \rightarrow M$ satisfying $\nabla_{\dot{\gamma}(t)} \dot{\gamma} = 0$ for $t \in [a, b]$. By a *unit-speed geodesic ray* we mean a geodesic $\gamma : [0, \infty) \rightarrow M$ with $\|\dot{\gamma}(0)\| = 1$ (and then $\|\dot{\gamma}(t)\| = 1$ for all t). For $x \in M$ and $u \in T_x$, there is a unique geodesic $\gamma(t) = \exp_x tu$ with $\gamma(0) = x$ and $\dot{\gamma}(0) = u$. By the completeness of M , the map $t \mapsto \exp_x tu$ is defined on $\mathbb{R}_+$. This gives rise to a surjective map $\exp_x : T_x \rightarrow M$, called the *exponential map*.

For a map $\varphi : M \rightarrow N$, where N is another manifold, let $d\varphi : T_x(M) \rightarrow T_{\varphi(x)}(N)$ denote the differential of φ at $x \in M$. The differential $df = df_x : T_x \rightarrow \mathbb{R}$ of a function $f : M \rightarrow \mathbb{R}$ is given by $df(u) = (d/dt)f(\gamma(t))|_{t=0}$, where γ is a path with $\gamma(0) = x$ and $\dot{\gamma}(0) = u \in T_x$. The gradient $\nabla f(x) \in T_x$ of f is then defined via

$$\langle \nabla f(x), u \rangle := df(u) \quad (u \in T_x).$$

The covariant differentiation of the gradient vector field ∇f gives rise to the *Hessian* $\nabla^2 f(x) : T_x \rightarrow T_x$:

$$\nabla^2 f(x)u := \nabla_u \nabla f(x) \quad (u \in T_x).$$

The Hessian is a symmetric operator in the sense that $\langle \nabla^2 f(x)u, v \rangle = \langle \nabla^2 f(x)v, u \rangle$.

Complex projective space. We will consider the complex projective space as a Riemannian manifold. Let V be an n -dimensional vector space over $\mathbb{C}$. The *complex projective space* $\mathbb{P}(V)$ over V is a quotient manifold $V \setminus \{0\} / \sim$ by the equivalent relation

$v \sim v' \Leftrightarrow v = \alpha v'$ for some nonzero $\alpha \in \mathbb{C} \setminus \{0\}$. The image of v by $V \setminus \{0\} \rightarrow \mathbb{P}(V)$ is denoted by $[v]$. A Riemannian structure on $\mathbb{P}(V)$ is given by the *Fubini-Study form* as follows. Let $(\cdot, \cdot)$ be a Hermitian inner product on V . Regard V as a $2n$ -dimensional Euclidean space by the real inner product $Re(\cdot, \cdot)$. This induces a Riemannian structure on the sphere $S^{2n-1} = \{v \in V \mid \|v\| = r\}$, where we set $r := \sqrt{2}$. Then $\mathbb{P}(V)$ is viewed as the Riemannian quotient of S^{2n-1} by isometry $(e^{i\theta}, v) \mapsto e^{i\theta}v$ for $e^{i\theta} \in U_1$. The resulting metric on $\mathbb{P}(V)$ is called the *Fubini-Study metric*. See e.g., [8, Chapter 9] for Riemannian quotient manifolds.

2.2 Hadamard manifolds

A *Hadamard manifold* M is simply-connected complete Riemannian manifold having nonpositive sectional curvature everywhere; see [46, V.4]. For any two points in M , a geodesic connecting them is uniquely determined (up to an affine rescaling). The exponential map $\exp_x$ is a diffeomorphism from T_x to M . The parallel transport from T_x to T_y along the geodesic is simply written as $\tau_{x \rightarrow y}$.

In this paper, the *boundary M^∞ at infinity* and the *cone topology* on $M \cup M^\infty$ play particularly important roles; see [46, V.4.2] for a quick introduction to these notions. Two unit-speed geodesic rays $\gamma, \gamma' : \mathbb{R}_+ \rightarrow M$ are called *asymptotic* if $d(\gamma(t), \gamma'(t)) < C$ ($t \in \mathbb{R}_+$) for some constant $C > 0$. The asymptotic relation is an equivalence relation on the set of all unit-speed geodesic rays. Let M^∞ denote the set of equivalence classes for all rays. Let us fix an arbitrary point $x \in M$. Any unit vector $u \in S_x$ defines an asymptotic class of unit-speed geodesic ray $t \mapsto \exp_x tu$. This correspondence is a bijection between S_x and M^∞ , and induces a topology on M^∞ that is isomorphic to sphere S_x . In fact, this topology is independent of the choice of x . Further, the topologies on M and on M^∞ are extended to $M \cup M^\infty$ as follows. Since $\exp_x$ is a diffeomorphism, it holds $M \simeq (S_x \times \mathbb{R}_+)/\sim_0$, where $\sim_0$ is an equivalence relation $(u, r) \sim_0 (u', r') \Leftrightarrow (u, r) = (u', r')$ or $r = r' = 0$. With $M^\infty \simeq S_x \times \{\infty\}$, we obtain compact Hausdorff space $M \cup M^\infty \simeq (S_x \times (\mathbb{R}_+ \cup \{\infty\}))/\sim_0$ (isomorphic to B_x). This topology on $M \cup M^\infty$ is called the *cone topology*. In this topology, a sequence x_i in M converges to $\xi \in M^\infty$ if and only if

- $d(x, x_i) \rightarrow \infty$, and
- the sequence u_i in S_x determined by $x_i = \exp_x d(x, x_i)u_i$ converges to $u \in S_x$, where the asymptotic class of geodesic $t \mapsto \exp_x tu$ is equal to ξ .

The *angle* $\angle^\infty(\xi, \xi')$ of two points $\xi, \xi' \in M^\infty$ is defined as $\sup_{x \in M} \angle(u, u')$, where u and u' are the representatives of ξ and ξ' , respectively, at T_x . The angle defines a metric on M^∞ , which induces a different topology. By using the angle metric on M^∞ , we can define a metric d^∞ on the *Euclidean cone* $CM^\infty := (M^\infty \times \mathbb{R}_+)/\sim_0$ of the boundary M^∞ by $d^\infty((\xi, r), (\xi', r'))^2 = r^2 + (r')^2 - 2rr' \cos \angle^\infty(\xi, \xi')$. This space CM^∞ is viewed as the space of asymptotic classes of (not necessarily unit-speed) geodesic rays. It is identified with T_x , though the metric space (CM^∞, d^∞) has a different topology from T_x and is not necessarily a manifold. The metric space (CM^∞, d^∞) is known as a *Hadamard space*—a complete geodesic metric space satisfying the *CAT(0)-inequality* [9]. It is uniquely geodesic, and its convexity is defined along geodesics. The unit ball $B^\infty = \{p \in CM^\infty \mid d^\infty(0, p) \leq 1\}$ around the origin 0 is a convex set, where the origin 0 is the image of point (r, ξ) with $r = 0$. Observe that B^∞ can be identified with B_x for any $x \in M$.

Manifold of positive definite matrices and symmetric space. A representative example of Hadamard manifolds is the space P_n of $n \times n$ positive definite Hermitian matrices; see [9, II. 10]. The tangent space T_x at $x \in P_n$ is identified with the vector space $\mathfrak{p}_n$ of Hermitian matrices, and the Riemannian metric is given by $\langle G, H \rangle_x := \text{tr } x^{-1} H x^{-1} G$. In this case, several manifold notions are explicitly written; see, e.g., [30, Section 5.2]. The exponential map $\exp_x$ at x is given by $H \mapsto x^{1/2} e^{x^{-1/2} H x^{-1/2}} x^{1/2}$, where $e^{\cdot}$ is the matrix exponential. Particularly, any geodesic issuing from x is of form $t \mapsto x^{1/2} e^{t x^{-1/2} H x^{-1/2}} x^{1/2}$ for some Hermitian matrix $H \in T_x$ with $\|H\| = \|x^{-1/2} H x^{-1/2}\|_F = 1$, where $\|\cdot\|_F$ is the Frobenius norm. An explicit formula of the geodesic parallel transport $\tau_{x \rightarrow y}$ is also known. We will use one special case: $\tau_{x \rightarrow I} H = x^{-1/2} H x^{-1/2}$.

Any totally geodesic subspace M of P_n is also a Hadamard manifold. Here a submanifold $M \subseteq P_n$ is said to be *totally geodesic* if every geodesic in M is also geodesic in P_n . It is known that for a connected Lie group $G \subseteq GL_n$ defined by polynomials and satisfying $G = G^\dagger$, submanifold $P_n \cap G$ is a totally geodesic subspace. Such a group G is called a *reductive algebraic group*, and $P_n \cap G$ is known as a *symmetric space* (of nonpositive curvature). A particular case we will face is: $G = SL_n$ and $P_n^1 := P_n \cap SL_n = \{x \in P_n \mid \det x = 1\}$, where tangent space $T_I(P_n^1)$ at I is given by $\mathfrak{p}_n^1 := \{H \in \mathfrak{p}_n \mid \text{tr } H = 0\}$. It is known that the boundary M^∞ at infinity of the symmetric space $P_n \cap G$ becomes *spherical building*, and the associated cone CM^∞ becomes a *Euclidean building*; see e.g., [9, II. 10]. We will consider convex functions on these spaces in Section 4.

2.3 Convex functions

In Hadamard manifold M , by the uniqueness of geodesics, convexity is naturally introduced. A function $f : M \rightarrow \mathbb{R}$ is said to be *convex* if for every geodesic $\gamma : [a, b] \rightarrow M$ one dimensional function $f \circ \gamma : [a, b] \rightarrow \mathbb{R}$ is convex. This condition is equivalent to $(f \circ \gamma)''(t) \geq 0$. From $(f \circ \gamma)''(t) = (d/dt) \langle \nabla f(\gamma(t)), \dot{\gamma}(t) \rangle = \langle \nabla_{\dot{\gamma}(t)} \nabla f(\gamma(t)), \dot{\gamma}(t) \rangle$, the convexity of f is equivalent to the positive semidefiniteness of the Hessian $\nabla^2 f(x)$:

$$\langle \nabla^2 f(x) u, u \rangle \geq 0.$$

for all $x \in M, u \in T_x$. We also consider the Lipschitz condition for gradient vector field ∇f . A function $f : M \rightarrow \mathbb{R}$ is said to be *L-smooth* with $L \in \mathbb{R}_+$ if

$$\langle \nabla^2 f(x) u, u \rangle \leq L \langle u, u \rangle$$

for all $x \in M, u \in T_x$. That is, the operator norm $\|\nabla^2 f(x)\|$ is bounded by L for all $x \in M$.

We next introduce an important tool for studying unboundedness of a convex function. Let us fix $x_0 \in M$. The *recession function (asymptotic slope)* $f^\infty : M^\infty \rightarrow \mathbb{R} \cup \{\infty\}$ [29, 36] is defined by

$$f^\infty(u) = f_{x_0}^\infty(u) := \lim_{s \rightarrow \infty} \frac{f(\exp_{x_0} s u) - f(x_0)}{s} = \lim_{s \rightarrow \infty} \frac{f(\exp_{x_0} s u)}{s} \quad (u \in S_{x_0} \simeq M^\infty), \quad (2.1)$$

where f^∞ is independent of x_0 [38, Lemma 2.10]: If $t \mapsto \exp_{x_0} t u$ and $t \mapsto \exp_{y_0} t v$ are asymptotic, then $f_{x_0}^\infty(u) = f_{y_0}^\infty(v)$.¹ The recession function f^∞ is naturally extended

¹Proof sketch: Let $\alpha(t) := \exp_{x_0} t u$ and $\beta(t) := \exp_{y_0} t v$, and define $u_t \in S_{x_0}$ by $\exp_{x_0} d(x_0, \beta(t)) u_t =$

to $CM^\infty \rightarrow \mathbb{R} \cup \{\infty\}$ by allowing u to any vector in $T_{x_0} \simeq CM^\infty$. If $M = \mathbb{R}^n$, then $CM^\infty = \mathbb{R}^n$ and f^∞ matches the recession function in Euclidean convex analysis; see [45, Section 8] and [31, Section 3.2]. As in the Euclidean case, the following properties hold:

$$\begin{aligned} \inf_{\xi \in M^\infty} f^\infty(\xi) < 0 &\Rightarrow \inf_{x \in M} f(x) = -\infty. \\ \inf_{\xi \in M^\infty} f^\infty(\xi) > 0 &\Rightarrow \exists x^* \in M : f(x^*) = \inf_{x \in M} f(x). \end{aligned} \quad (2.2)$$

The second property is included in [36, Lemma 3.2 (vi)]. Moreover, it is known [29] that f^∞ is a positively homogeneous convex function on Hadamard space CM^∞ .

In particular, both $\inf_{\xi \in M^\infty} f^\infty(\xi) < 0$ and $\inf_{x \in M} \|\nabla f(x)\| > 0$ are sufficient conditions for unboundedness of f . In fact, they are equivalent.

Proposition 2.1 ([36, Lemma 3.2 (iii), Lemma 3.4]; see also [29]).

- (1) $\inf_{\xi \in M^\infty} f^\infty(\xi) < 0$ if and only if $\inf_{x \in M} \|\nabla f(x)\| > 0$.
- (2) If $\inf_{\xi \in M^\infty} f^\infty(\xi) < 0$, then there uniquely exists $\xi^* \in M^\infty$ with $f^\infty(\xi^*) = \inf_{\xi \in M^\infty} f^\infty(\xi)$.

The existence in (2) follows from lower-semicontinuity of f^∞ on the compact space M^∞ with respect to the cone topology. The uniqueness of ξ^* in (2) can be seen from positively homogeneous convexity of f^∞ on CM^∞ , as in the Euclidean case.²

As a sharpening of the easier part (the only-if part) in (1), we here mention the following weak duality between the gradient-norm and the recession function.

Lemma 2.2 (Weak duality). $\inf_{x \in M} \|\nabla f(x)\| \geq \sup_{\xi \in B^\infty} -f^\infty(\xi)$.

Proof. For $x \in M$ and $\xi \in B_x \simeq B^\infty$, it holds

$$f^\infty(\xi) = \lim_{t \rightarrow \infty} \frac{f(\exp_x t\xi) - f(x)}{t} \geq \lim_{t \rightarrow 0} \frac{f(\exp_x t\xi) - f(x)}{t} = \langle \nabla f(x), \xi \rangle \geq -\|\nabla f(x)\|.$$

□

In Section 3, we show, via the gradient flow of f , that the equality (*strong duality*) always holds. This technique may be viewed as a refinement of the proof of the if-part in Proposition 2.1 (1) in [36], in which a limit of the gradient flow of f constructs ξ with $f^\infty(\xi) < 0$. A similar gradient-flow approach can be found in [15, 24, 50] for the setting of GIT; see Section 4.1.

$\beta(t)$. By convexity of f along geodesic between x_0 and $\beta(t)$, it holds $f(\exp_{x_0} su_t) - f(x_0) \leq (s/d(x_0, \beta(t)))(f(\beta(t)) - f(x_0))$ for $s \in [0, d(x_0, \gamma(t))]$. By triangle inequality, we have $(f(\beta(t)) - f(x_0))/d(x_0, \beta(t)) \leq \max_{\sigma \in \{-1, 1\}} (f(\beta(t)) - f(x_0))/(t + \sigma d(x_0, y_0)) \rightarrow f_{y_0}^\infty(v)$ for $t \rightarrow \infty$. By CAT(0)-inequality on the geodesic triangle of vertices $x_0, \alpha(t), \beta(t)$ and by $d(\alpha(t), \beta(t))$ bounded, it holds $\exp_{x_0} su_t \rightarrow \alpha(s)$ for $t \rightarrow \infty$. Thus we have $(f(\alpha(s)) - f(x_0))/s \leq f_{y_0}^\infty(v)$ and $f_{x_0}^\infty(u) \leq f_{y_0}^\infty(v)$, and $f_{x_0}^\infty(u) = f_{y_0}^\infty(v)$ by symmetry.

²If $f^\infty(\xi) = f^\infty(\xi') = c < 0$, then by convexity, it holds $f^\infty(m) \leq (f^\infty(\xi) + f^\infty(\xi'))/2 = c$ for the midpoint m of ξ and ξ' in CM^∞ , and by $\|m\| < 1$ it holds $f^\infty(m/\|m\|) = f^\infty(m)/\|m\| < c$.

3 Asymptotic behavior of gradient flow

3.1 Continuous-time gradient flow

Let $f : M \rightarrow \mathbb{R}$ be a convex function. Consider the following differential equation—the *gradient flow* of f ,

$$\frac{dx(t)}{dt} = -\nabla f(x(t)), \quad x(0) = x_0. \quad (3.1)$$

It is clear that the trajectory $x(t)$ is going to minimize f ; see Lemma 3.2 (2) below. In fact, if a minimizer of f exists, then $x(t)$ converges to a minimizer. This convergence is known for a general setting of Hadamard spaces; see e.g., [5, Theorem 5.1.16] and [40, Theorem 2.41]. Our focus is the case where f is unbounded below, particularly the case where the minimum gradient-norm is positive. We establish the following convergence of an unbounded gradient flow and the strong duality between the gradient-norm and the recession function.

Theorem 3.1. *Suppose that $\kappa^* := \inf_{x \in M} \|\nabla f(x)\| > 0$. Let $x(t)$ be the solution of (3.1).*

- (1) $\|\nabla f(x(t))\|$ converges to the minimum gradient norm κ^* , and
- (2) $x(t)$ converges, in cone topology, to the unique minimizer ξ^* of f^∞ over M^∞ ,

where the following equality holds

$$\lim_{t \rightarrow \infty} \|\nabla f(x(t))\| = \inf_{x \in M} \|\nabla f(x)\| = \sup_{\xi \in M^\infty} -f^\infty(\xi) = -f^\infty\left(\lim_{t \rightarrow \infty} x(t)\right). \quad (3.2)$$

We should mention related results. In a general setting of a Hadamard space X , Caprace and Lytchak [14, Proposition 4.2] showed that the gradient-flow curve of a Lipschitz convex function with $\kappa^* > 0$ converges to a point in the boundary X^∞ of X . Their proof relies on a very general result of Karlsson and Margulis [37, Theorem 2.1] for semi-contraction semigroups in uniformly convex spaces. Here it is well-known³ that the gradient-flow semigroup ϕ_t satisfies the *(semi-)contraction property*:

$$d(\phi_t(x), \phi_t(y)) \leq d(x, y) \quad (t \in \mathbb{R}_+, x, y \in M), \quad (3.3)$$

where $\phi_t(x)$ is the solution of (3.1) with initial point $x(0) = x$. If the *velocity of escape*

$$\kappa^*(x) := \limsup_{t \rightarrow \infty} \frac{d(\phi_t(x), x)}{t} \quad (3.4)$$

is positive, then [37, Theorem 2.1] is applicable for the convergence of $\phi_t(x)$ in M^∞ ; Caprace and Lytchak actually showed that $\kappa^* > 0$ implies $\kappa^*(x) > 0$. Although with more effort one can deduce the whole statement of Theorem 3.1, we take a different approach which relies on neither [37] nor contraction property (3.3). As mentioned after Lemma 2.2, our proof is partly inspired by an idea in [36] but directly establishes the relation (3.2). An advantage of this approach is enable to adapt to the discrete setting in Section 3.2.

We start with the following well-known properties of gradient flows.

³For a general setting of gradient flows in metric spaces, it is found in [3, Theorem 4.0.4]. For our manifold case, it is an easy consequence of the first variation formula [46, Proposition 2.2] as follows: $(d/dt)d(\phi_t(x), \phi_t(y))^2/2 = \langle -\nabla f(\phi_t(y)), \dot{\gamma}(1) \rangle - \langle -\nabla f(\phi_t(x)), \dot{\gamma}(0) \rangle = -\int_0^1 (d/dt)\langle \nabla f(\gamma(s)), \gamma(s) \rangle ds = -\int_0^1 \langle \nabla^2 f(\gamma(s))\dot{\gamma}(s), \dot{\gamma}(s) \rangle ds \leq 0$, where $\gamma : [0, 1] \rightarrow M$ is a geodesic from $\phi_t(x)$ to $\phi_t(y)$.

Lemma 3.2. (1) The solution $x(t)$ of (3.1) is defined on $\mathbb{R}_+$.

(2) $t \mapsto f(x(t))$ is nonincreasing.

(3) $t \mapsto \|\nabla f(x(t))\|$ is nonincreasing.

We describe a proof since the intermediate equations will be used.

Proof. (2) follows from $(d/dt)f(x(t)) = \langle \nabla f(x(t)), \dot{x}(t) \rangle = -\|\nabla f(x(t))\|^2 \leq 0$.

(3) follows from $(d/dt)\|\nabla f(x(t))\|^2 = -2\langle \nabla^2 f(x(t))\dot{x}(t), \dot{x}(t) \rangle \leq 0$.

(1). Suppose that $x(t)$ is defined on $[0, T)$ for finite $T > 0$. The length ℓ of curve $x(t)$ is given by

$$\ell = \int_0^T \|\dot{x}(t)\| dt = \int_0^T \|\nabla f(x(t))\| dt \leq \|\nabla f(x_0)\| T,$$

where the inequality follows from (3). This means that the curve belongs to the closed ball B of radius $\|\nabla f(x_0)\|T$ with center x_0 . Then, $x(t)$ is connected to the solution of (3.1) with initial point $\lim_{t \rightarrow T} x(t) \in B$. Thus the solution of (3.1) is defined on $\mathbb{R}_+$. $\square$

Proof of Theorem 3.1. Let $\kappa := \lim_{t \rightarrow \infty} \|\nabla f(x(t))\| \geq \kappa^*$. First, we note

$$f(x(t)) - f(x_0) = \int_0^t \frac{d}{d\tau} f(x(\tau)) d\tau = - \int_0^t \|\nabla f(x(\tau))\|^2 d\tau \leq -\kappa^2 t, \quad (3.5)$$

$$d(x(t), x_0) \leq \int_0^t \|\dot{x}(\tau)\| d\tau = \int_0^t \|\nabla f(x(\tau))\| d\tau. \quad (3.6)$$

Then it holds $d(x(t), x_0) \rightarrow \infty$. Otherwise, $x(t)$ has an accumulation point x^* in M and $f(x^*) = -\infty$ by (3.5), contradicting $f(x^*) \in \mathbb{R}$.

Define $u(t) \in S_{x_0}$ via $x(t) = \exp_{x_0} d(x(t), x_0)u(t)$. For $s \in (0, d(x(t), x_0)]$, by convexity of f along geodesic from x_0 to $x(t)$, it holds

$$f(\exp_{x_0} su(t)) - f(x_0) \leq \frac{s}{d(x(t), x_0)} (f(x(t)) - f(x_0)).$$

From this, we have

$$\begin{aligned} \frac{f(\exp_{x_0} su(t)) - f(x_0)}{s} &\leq \frac{f(x(t)) - f(x_0)}{d(x(t), x_0)} \leq - \frac{\int_0^t \|\nabla f(x(\tau))\|^2 d\tau}{\int_0^t \|\nabla f(x(\tau))\| d\tau} \\ &\leq -\frac{1}{t} \int_0^t \|\nabla f(x(\tau))\| d\tau \leq -\kappa, \end{aligned}$$

where the second inequality follows from (3.5) and (3.6), the third from the Cauchy-Schwartz inequality $(\int_0^t F(\tau)G(\tau)d\tau)^2 \leq \int_0^t F(\tau)^2 d\tau \int_0^t G(\tau)^2 d\tau$ for $F(\tau) := \|\nabla f(x(\tau))\|$ and $G(\tau) := 1$, and the fourth from Lemma 3.2 (3).

Choose any convergence subsequence $u(t_i)$ with $t_i \rightarrow \infty$ and $u(t_i) \rightarrow u^*$. Then it holds

$$\frac{f(\exp_{x_0} su^*) - f(x_0)}{s} \leq -\kappa.$$

For $s \rightarrow \infty$, we have $f^\infty(u^*) \leq -\kappa$. Then, by weak duality (Lemma 2.2),

$$\inf_{\xi \in M^\infty} f^\infty(\xi) \leq f^\infty(u^*) \leq -\kappa \leq -\kappa^* = \sup_{x \in M} -\|\nabla f(x)\| \leq \inf_{\xi \in M^\infty} f^\infty(\xi).$$

Thus the equality holds. Since the minimum ξ^* of f^∞ over M^∞ uniquely exists (Proposition 2.1 (2)), it must hold $\xi^* = u^*$. Any convergence subsequence $u(t_i)$ of $u(t)$ converges to ξ^* . Since S_{x_0} is compact, $u(t)$ itself converges to ξ^* . $\square$

Even if $\kappa^* = 0$, the strong duality holds (since $f^\infty(0) = 0$):

Corollary 3.3. $\inf_{x \in M} \|\nabla f(x)\| = \sup_{\xi \in B^\infty} -f^\infty(\xi)$.

The velocity of escape (3.4) coincides with the minimum gradient-norm.

Proposition 3.4. Suppose that $\kappa^* = \inf_{x \in M} \|\nabla f(x)\| > 0$. Then it holds

$$\lim_{t \rightarrow \infty} \frac{d(x_0, x(t))}{t} = \kappa^*.$$

Proof. For $t > s \geq 0$, it holds $d(x(s), x(t)) \leq \int_s^t \|\nabla f(x(\tau))\| d\tau \leq \|\nabla f(x(s))\|(t - s)$. Hence

$$\limsup_{t \rightarrow \infty} \frac{d(x_0, x(t))}{t} = \limsup_{t \rightarrow \infty} \frac{d(x(s), x(t))}{t - s} \leq \|\nabla f(x(s))\|.$$

On the other hand, we have

$$\begin{aligned} -\|\nabla f(x(t))\|^2(t - s) &\geq -\int_s^t \|\nabla f(x(\tau))\|^2 d\tau = f(x(t)) - f(x(s)) \\ &\geq \langle \dot{\gamma}(0), \nabla f(x(s)) \rangle d(x(s), x(t)) \geq -\|\nabla f(x(s))\| d(x(s), x(t)), \end{aligned}$$

where γ is the unit speed geodesic from $x(s)$ to $x(t)$. Thus it holds

$$\begin{aligned} \liminf_{t \rightarrow \infty} \frac{d(x_0, x(t))}{t} &\geq \liminf_{t \rightarrow \infty} \frac{d(x(t), x(s)) - d(x_0, x(s))}{t} = \liminf_{t \rightarrow \infty} \frac{d(x(t), x(s))}{t - s} \\ &\geq \frac{\lim_{t \rightarrow \infty} \|\nabla f(x(t))\|^2}{\|\nabla f(x(s))\|} = \frac{(\kappa^*)^2}{\|\nabla f(x(s))\|}. \end{aligned}$$

For arbitrary large s , $\|\nabla f(x(s))\|$ can be arbitrary close to κ^* . Hence we have

$$\kappa^* \leq \liminf_{t \rightarrow \infty} \frac{d(x_0, x(t))}{t} \leq \limsup_{t \rightarrow \infty} \frac{d(x_0, x(t))}{t} \leq \kappa^*.$$

$\square$

We next consider the “convergence” of gradient $\nabla f(x(t))$. Since the space $T_{x(t)}$ varies, the convergence concept of $\nabla f(x(t))$ is less obvious. In our intuition, $\nabla f(x(t))$ and ξ^* would have opposite directions in limit. The following partially justifies this intuition.

Proposition 3.5. Suppose that $\kappa^* := \inf_{x \in M} \|\nabla f(x)\| > 0$. Let $\xi^* \in S_{x_0}$ denote the representative of the unique minimizer of f^∞ over $M^\infty \simeq S_{x_0}$. Then it holds

$$\liminf_{t \rightarrow \infty} \|\tau_{x(t) \rightarrow x_0} \nabla f(x(t)) + \kappa^* \xi^*\| = 0.$$

Question 3.6. Does $\lim_{t \rightarrow \infty} \tau_{x(t) \rightarrow x_0} \nabla f(x(t)) = -\kappa^* \xi^*$ hold?

We will see in Section 4 that this property holds for the Kempf-Ness function and has important consequences.

Proof of Proposition 3.5. Let γ_t be the unit-speed geodesic from x_0 to $x(t)$. Let $d(t) := d(x_0, x(t))$. Then, by [46, Chapter III, Proposition 4.8 (1)], it holds $d(t)' = \langle \dot{\gamma}_t(d(t)), \dot{x}(t) \rangle$. Therefore, we have

$$\limsup_{t \rightarrow \infty} d(t)' = \limsup_{t \rightarrow \infty} \langle \dot{\gamma}_t(d(t)), \dot{x}(t) \rangle \leq \lim_{t \rightarrow \infty} \|\dot{x}(t)\| = \lim_{t \rightarrow \infty} \|\nabla f(x(t))\| = \kappa^*. \quad (3.7)$$

On the other hand, by Proposition 3.4, it holds $\kappa^* = \limsup_{t \rightarrow \infty} d(t)/t \leq \limsup_{t \rightarrow \infty} d(t)'$, where the inequality is a general property. Thus the equality holds in (3.7). Necessarily we have

$$\limsup_{t \rightarrow \infty} \angle(\dot{\gamma}_t(d(t)), \nabla f(x(t))) = \pi. \quad (3.8)$$

By $\|\nabla f(x(t))\| \rightarrow \kappa^*$, we have $\liminf_{t \rightarrow \infty} \|\nabla f(x(t)) + \kappa^* \dot{\gamma}_t(d(t))\| = 0$. With parallel transport $\tau_{x(t) \rightarrow x_0}$ and $\dot{\gamma}_t(0) \rightarrow \xi^*$, we have the claim. $\square$

3.2 Discrete-time gradient flow (gradient descent)

Next we consider the discrete version. Suppose that $f : M \rightarrow \mathbb{R}$ is an L -smooth convex function. Consider the following sequence:

$$x_{i+1} := \exp_{x_i} \left(-\frac{1}{L} \nabla f(x_i) \right) \quad (i = 0, 1, \dots). \quad (3.9)$$

This is nothing but the trajectory generated by the gradient descent with initial point x_0 and step-size $1/L$; we remark in Remark 3.12 another type of discrete gradient flow. The convergence/accumulation of x_i to a minimizer of f can be shown under several reasonable assumptions; see e.g., [8, Theorem 11.29]. For an unbounded case, as in the continuous setting, we establish the following.

Theorem 3.7. *Suppose that $\kappa^* := \inf_{x \in M} \|\nabla f(x)\| > 0$. Let x_i be the sequence in (3.9).*

- (1) $\|\nabla f(x_i)\|$ converges to the minimum gradient norm κ^* , and
- (2) x_i converges, in cone topology, to the unique minimizer $\xi^* \in M^\infty$ of f^∞ .

Hence the following holds

$$\lim_{i \rightarrow \infty} \|\nabla f(x_i)\| = \inf_{x \in M} \|\nabla f(x)\| = \sup_{\xi \in M^\infty} -f^\infty(\xi) = -f^\infty \left(\lim_{i \rightarrow \infty} x_i \right). \quad (3.10)$$

Our original attempt proving this was to establish the contraction property

$$d(\phi_i(x), \phi_i(y)) \leq d(x, y) \quad (x, y \in M, i = 1, 2, \dots), \quad (3.11)$$

for the semigroup ϕ_i of (3.9), and to apply the approach of [14, 37]. But we could not do it, and do not know whether (3.11) is true. Note that (3.11) is true in Euclidean space $M = \mathbb{R}^n$; see e.g., [47, Example 1].

The proof goes a way analogous to Theorem 3.1. Corresponding to Lemma 3.2, the following properties hold.

Lemma 3.8. (1) $f(x_{i+1}) \leq f(x_i) - \frac{1}{L} \|\nabla f(x_{i+1})\|^2$.

(2) $\|\nabla f(x_{i+1})\| \leq \|\nabla f(x_i)\|$.

Contrary to the well-known inequality $f(x_{i+1}) \leq f(x_i) - (1/2L)\|\nabla f(x_i)\|^2$ (see [8, (11.15)]), our inequality (1) seems less well-known; see Remark 3.13 for further discussion.

Proof. (2). Let $\gamma(t) := \exp_{x_i} -t\nabla f(x_i)$. Then we have

$$\begin{aligned}\tau_\gamma^{-1/L}\nabla f(x_{i+1}) &= \nabla f(x_i) + \int_0^{1/L} \frac{d}{ds} \tau_\gamma^{-s} \nabla f(\gamma(s)) ds \\ &= \nabla f(x_i) + \int_0^{1/L} \tau_\gamma^{-s} \nabla_{\dot{\gamma}(s)} \nabla f(\gamma(s)) ds \\ &= L \int_0^{1/L} \left(I - \frac{1}{L} \tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s \right) \nabla f(x_i) ds,\end{aligned}$$

where we use $\dot{\gamma}(s) = -\tau_\gamma^s \nabla f(x_i)$ since γ is a geodesic. Since $\langle \cdot, \cdot \rangle$ is invariant under parallel transport, the operator norm of $\tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s$ is equal to that of $\nabla^2 f(\gamma(s))$. By convexity and L -smoothness, all eigenvalues of $\tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s$ belong to $[0, L]$. Hence we have

$$\begin{aligned}\|\nabla f(x_{i+1})\| &= \|\tau_\gamma^{-1/L} \nabla f(x_{i+1})\| \leq L \int_0^{1/L} \left\| I - \frac{1}{L} \tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s \right\| \|\nabla f(x_i)\| ds \\ &\leq \|\nabla f(x_i)\|.\end{aligned}$$

(1).

$$\begin{aligned}\left\| \tau_\gamma^{-1/L} \nabla f(x_{i+1}) - \frac{1}{2} \nabla f(x_i) \right\| &= L \left\| \int_0^{1/L} \left(\frac{1}{2} I - \frac{1}{L} \tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s \right) \nabla f(x_i) ds \right\| \\ &\leq L \int_0^{1/L} \left\| \left(\frac{1}{2} I - \frac{1}{L} \tau_\gamma^{-s} \nabla^2 f(\gamma(s)) \tau_\gamma^s \right) \right\| \|\nabla f(x_i)\| ds \leq \frac{1}{2} \|\nabla f(x_i)\|.\end{aligned}$$

By squaring this, we have $\|\tau_\gamma^{-1/L} \nabla f(x_{i+1})\|^2 \leq \langle \tau_\gamma^{-1/L} \nabla f(x_{i+1}), \nabla f(x_i) \rangle$, particularly,

$$\|\nabla f(x_{i+1})\|^2 \leq \langle \nabla f(x_{i+1}), \tau_\gamma^{1/L} \nabla f(x_i) \rangle. \quad (3.12)$$

From convexity, it holds

$$\begin{aligned}f(x_i) &\geq f(x_{i+1}) + \frac{1}{L} \frac{d}{dt} f(\gamma(1/L - t)) \big|_{t=0} = f(x_{i+1}) - \frac{1}{L} \langle \nabla f(x_{i+1}), \dot{\gamma}(1/L) \rangle \\ &= f(x_{i+1}) + \frac{1}{L} \langle \nabla f(x_{i+1}), \tau_\gamma^{1/L} \nabla f(x_i) \rangle \geq f(x_{i+1}) + \frac{1}{L} \|\nabla f(x_{i+1})\|^2.\end{aligned}$$

□

Proof of Theorem 3.7. Let $\kappa := \lim_{i \rightarrow \infty} \|\nabla f(x_i)\| \geq \kappa^*$. For $i > 0$, we have

$$f(x_i) - f(x_0) \leq -\frac{1}{L} \sum_{k=1}^i \|\nabla f(x_k)\|^2 \leq -\frac{i}{L} \kappa, \quad (3.13)$$

$$d(x_i, x_0) \leq \sum_{k=0}^{i-1} d(x_k, x_{k+1}) = \frac{1}{L} \sum_{k=0}^{i-1} \|\nabla f(x_k)\|, \quad (3.14)$$

where the first inequality follows from Lemma 3.8 (1) and the second from the triangle inequality. Then $d(x_i, x_0) \rightarrow \infty$ holds (as before).

Let $u_i \in S_{x_0}$ be defined via $x_i = \exp_{x_0} d(x_i, x_0)u_i$. For $s \in (0, d(x_i, x_0)]$, by convexity of f along $[x_i, x_0]$ it holds

$$f(\exp_{x_0} su_i) - f(x_0) \leq \frac{s}{d(x_i, x_0)}(f(x_i) - f(x_0)).$$

From this, we have

$$\begin{aligned} \frac{f(\exp_{x_0} su_i) - f(x_0)}{s} &\leq \frac{f(x_0) - f(x_i)}{d(x_i, x_0)} \leq -\frac{\sum_{k=1}^i \|\nabla f(x_k)\|^2}{\sum_{k=0}^{i-1} \|\nabla f(x_k)\|} \\ &= -\frac{\sum_{k=0}^{i-1} \|\nabla f(x_k)\|^2}{\sum_{k=0}^{i-1} \|\nabla f(x_k)\|} + \frac{\|\nabla f(x_0)\|^2 - \|\nabla f(x_i)\|^2}{\sum_{k=0}^{i-1} \|\nabla f(x_k)\|} \\ &\leq -\frac{1}{i} \sum_{k=0}^{i-1} \|\nabla f(x_k)\| + \frac{\|\nabla f(x_0)\|^2}{\sum_{k=0}^{i-1} \|\nabla f(x_k)\|} \leq -\kappa + \frac{1}{i} \frac{\|\nabla f(x_0)\|^2}{\kappa}, \end{aligned} \quad (3.15)$$

where the second follows from (3.13) and (3.14), the third from the Cauchy-Schwarz inequality $(\sum_k F_k G_k)^2 \leq \sum_k F_k^2 \sum_k G_k^2$, and the fourth from Lemma 3.8 (2).

Choose any convergent subsequence $\{u_{i_k}\}$ of $\{u_i\}$, which converges to $u^* \in S_{x_0}$. The second term of (3.15) vanishes as $i_k \rightarrow \infty$. Then it holds

$$\frac{f(\exp_{x_0} su^*) - f(x_0)}{s} \leq -\kappa.$$

By $s \rightarrow \infty$, we have $f^\infty(u^*) \leq -\kappa$. The rest is the same as the last part of the proof of Theorem 3.1. $\square$

We show the limiting behavior of the decrement of the function-value, the change of gradients, and the velocity of escape.

Proposition 3.9. *Suppose that $\kappa^* := \inf_{x \in M} \|\nabla f(x)\| > 0$.*

$$(1) \lim_{i \rightarrow \infty} f(x_{i+1}) - f(x_i) = -\frac{(\kappa^*)^2}{L}.$$

$$(2) \lim_{i \rightarrow \infty} \|\tau_{x_i \rightarrow x_{i+1}} \nabla f(x_i) - \nabla f(x_{i+1})\| = 0.$$

$$(3) \lim_{i \rightarrow \infty} \frac{d(x_0, x_i)}{i} = \frac{\kappa^*}{L}.$$

Proof. (1). By convexity and Lemma 3.8 (1), we have

$$-\frac{1}{L} \|\nabla f(x_i)\|^2 \leq f(x_{i+1}) - f(x_i) \leq -\frac{1}{L} \|\nabla f(x_{i+1})\|^2.$$

By $i \rightarrow \infty$ with Theorem 3.7, we have the claim.

(2). The inequality (3.12) is also written as

$$\|\nabla f(x_{i+1})\|^2 \leq \|\nabla f(x_i)\| \|\nabla f(x_{i+1})\| \cos \angle(\nabla f(x_{i+1}), \tau_{x_i \rightarrow x_{i+1}} \nabla f(x_i)).$$

By $\|\nabla f(x_i)\| \rightarrow \kappa^*$, we have $\angle(\nabla f(x_{i+1}), \tau_{x_i \rightarrow x_{i+1}} \nabla f(x_i)) \rightarrow 0$, and the claim.

(3). By the triangle inequality, it holds $d(x_0, x_i) \leq \sum_{k=0}^{i-1} d(x_k, x_{k+1}) = \sum_{k=0}^{i-1} \|\nabla f(x_k)\|/L$. Hence

$$\limsup_{i \rightarrow \infty} \frac{d(x_0, x_i)}{i} \leq \frac{1}{L} \limsup_{k \rightarrow \infty} \|\nabla f(x_k)\| = \frac{\kappa^*}{L}.$$

On the other hand, for arbitrary $0 \leq i < j$, we have

$$\begin{aligned} -\frac{j-i}{L} \|\nabla f(x_j)\|^2 &\geq -\frac{1}{L} \sum_{k=i+1}^j \|\nabla f(x_k)\|^2 \geq (f(x_j) - f(x_i)) \\ &\geq \langle \dot{\gamma}(0), \nabla f(x_i) \rangle d(x_i, x_j) \geq -\|\nabla f(x_i)\| d(x_i, x_j), \end{aligned}$$

where the first inequality follows from Lemma 3.8 (2), the second from (1), and the third from the convexity of f , and γ is the unit speed geodesic from x_i to x_j . Thus for arbitrary $i \geq 0$, it holds

$$\begin{aligned} \liminf_{j \rightarrow \infty} \frac{d(x_0, x_j)}{j} &\geq \liminf_{j \rightarrow \infty} \frac{d(x_i, x_j) - d(x_0, x_i)}{j} = \liminf_{j \rightarrow \infty} \frac{d(x_i, x_j)}{j-i} \\ &\geq \frac{1}{L} \liminf_{j \rightarrow \infty} \frac{\|\nabla f(x_j)\|^2}{\|\nabla f(x_i)\|} = \frac{1}{L} \frac{(\kappa^*)^2}{\|\nabla f(x_i)\|}. \end{aligned}$$

For arbitrary large i , $\|\nabla f(x_i)\|$ can be arbitrary close to κ^* . Hence we have

$$\frac{\kappa^*}{L} \leq \liminf_{i \rightarrow \infty} \frac{d(x_0, x_i)}{i} \leq \limsup_{i \rightarrow \infty} \frac{d(x_0, x_i)}{i} \leq \frac{\kappa^*}{L}.$$

□

For convergence of $\nabla f(x_i)$, the same property of Proposition 3.5 holds:

Proposition 3.10. *Suppose that $\kappa^* := \inf_{x \in M} \|\nabla f(x)\| > 0$. Let $\xi^* \in S_{x_0}$ denote the representative of the unique minimizer of f^∞ over $M^\infty \simeq S_{x_0}$. Then it holds*

$$\liminf_{t \rightarrow \infty} \|\tau_{x_i \rightarrow x_0} \nabla f(x_i) + \kappa^* \xi^*\| = 0.$$

Question 3.11. *Does $\lim_{i \rightarrow \infty} \tau_{x_i \rightarrow x_0} \nabla f(x_i) = -\kappa^* \xi^*$ hold?*

Proof of Proposition 3.10. Let $d_i := d(x_0, x_i)$. We first show

$$\limsup_{i \rightarrow \infty} d_{i+1} - d_i = \kappa^*/L. \quad (3.16)$$

Indeed, by the triangle inequality and Theorem 3.7 (1), we have $\limsup_{i \rightarrow \infty} d_{i+1} - d_i \leq \limsup_{i \rightarrow \infty} d(x_i, x_{i+1}) = \limsup_{i \rightarrow \infty} \|\nabla f(x_i)\|/L = \kappa^*/L$. On the other hand, by Proposition 3.9 (3), it holds $\kappa^*/L = \limsup_{i \rightarrow \infty} d_i/i \leq \limsup_{i \rightarrow \infty} d_{i+1} - d_i$, where the inequality is true for an arbitrary sequence.

Consider the geodesic triangle of vertices x_0, x_{i-1}, x_i . Let γ_i denote the unit-speed geodesic from x_0 to x_i . Let θ_i denote the angle at vertex x_i of this triangle. Then

$$\theta_i = \angle(\gamma_i(d_i), -\tau_{x_{i-1} \rightarrow x_i} \nabla f(x_{i-1})).$$

By the law of cosines in CAT(0) space M (see e.g., [9, II.1.9 (2)]), we have

$$\cos \theta_i \geq \frac{d_i^2 + d(x_{i-1}, x_i)^2 - d_{i-1}^2}{2d_i d(x_{i-1}, x_i)} = \frac{d(x_{i-1}, x_i)}{2d_i} + \frac{1}{2} \left(1 + \frac{d_{i-1}}{d_i}\right) \frac{d_i - d_{i-1}}{d(x_{i-1}, x_i)}.$$

Take $\limsup_{i \rightarrow \infty}$ in this inequality. By $d_i \rightarrow \infty$, $d(x_{i-1}, x_i) \rightarrow \kappa^*/L$, $d_{i-1}/d_i \rightarrow 1$ (seen from Proposition 3.9(3)), and (3.16), we have $\limsup_{i \rightarrow \infty} \cos \theta_i \geq 1$, and $\liminf_{i \rightarrow \infty} \theta_i = 0$. By Proposition 3.9 (2), it holds $\angle(\nabla f(x_i), \tau_{x_{i-1} \rightarrow x_i} \nabla f(x_{i-1})) \rightarrow 0$ and

$$\limsup_{i \rightarrow \infty} \angle(\dot{\gamma}_i(d_i), \nabla f(x_i)) = \pi.$$

By taking parallel transport $\tau_{x_i \rightarrow x_0}$ and $\dot{\gamma}_i(0) \rightarrow \xi^*$, we have the claim. □

Remark 3.12. Another type of discrete gradient flow, well-studied in the literature of nonpositively-curved space (see [5, 40, 44]), is defined via the *resolvent map* $J_\lambda^f : M \rightarrow M$,

$$J_\lambda^f(x) := \operatorname{argmin}_{y \in M} f(y) + \frac{1}{2\lambda} d(x, y)^2 \quad (x \in M), \quad (3.17)$$

where λ is a positive parameter. Let λ_i be a sequence of positive reals (satisfying $\lambda_i \rightarrow 0$ and $\sum_i \lambda_i \rightarrow \infty$). Then a discrete analogue (the *proximal point method*) of gradient flow is as follows:

$$x_{i+1} = J_{\lambda_i}^f(x_i) \quad (i = 0, 1, \dots). \quad (3.18)$$

For our manifold case, it can be written as an implicit difference scheme:

$$x_i = \exp_{x_{i+1}} \left(-\frac{1}{\lambda_i} \nabla f(x_{i+1}) \right). \quad (3.19)$$

Several nice (convergence) properties are known for the sequence of (3.18). For example, contraction property (3.11) holds for the semigroup of (3.18); see [5, Theorem 2.2.23]. On the other hand, solving (3.17) is a nontrivial task from algorithmic point of view

Remark 3.13. In the case of $M = \mathbb{R}^n$, Lemma 3.8 (1) can be easily obtained from a known inequality. For an L -smooth convex function f in $\mathbb{R}^n$, the following inequality holds (e.g., [6, Theorem 5.8 (iii)]):

$$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle + \frac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2 \quad (x, y \in \mathbb{R}^n),$$

though we do not know a reasonable manifold version to hold. By substituting $x = x_{i+1}$, $y = x_i$, and using $x_i - x_{i+1} = \nabla f(x_i)/L$, we have the following:

$$f(x_{i+1}) \leq f(x_i) - \frac{1}{2L} (\|\nabla f(x_i)\|^2 + \|\nabla f(x_{i+1})\|^2).$$

By $\|\nabla f(x_i)\| \geq \|\nabla f(x_{i+1})\|$ (Lemma 3.8 (2)), we have Lemma 3.8 (1).

3.3 Euclidean specialization

Here we present refinements of the above results for Euclidean setting $M = \mathbb{R}^n$. As far as our knowledge, the convergence results on discrete/continuous gradient flows seem new even in the Euclidean case, which are further sharpened as follows. In the Euclidean space $M = \mathbb{R}^n$, the tangent space T_x is also identified with $\mathbb{R}^n$ for every $x \in M$, where the inner product is given by $\langle u, v \rangle := u^\top v$. Notice that the parallel transport τ_γ for any path γ is the identity map. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. We assume the L -smoothness of f when the discrete gradient flow (3.9) is considered. The gradient $\nabla f(x) \in \mathbb{R}^n$ and Hessian $\nabla^2 f(x) \in \mathbb{R}^{n \times n}$ are obtained by $(\nabla f(x))_i = (\partial/\partial x_i)f(x)$ and $(\nabla^2 f(x))_{ij} = (\partial^2/\partial x_i \partial x_j)f(x)$, respectively. Let $f^* : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ be the Legendre-Fenchel conjugate of f :

$$f^*(p) := \sup\{\langle p, x \rangle - f(x) \mid x \in \mathbb{R}^n\} \quad (p \in \mathbb{R}^n).$$

Then it holds that

$$\overline{\nabla f(\mathbb{R}^n)} = \overline{\{p \in \mathbb{R}^n \mid f^*(p) < \infty\}} = \{p \in \mathbb{R}^n \mid \langle u, p \rangle \leq f^\infty(u) \ (u \in \mathbb{R}^n)\}. \quad (3.20)$$

Note that the duality relation (Corollary 3.3) can be seen from this expression; see the proof of Theorem 3.14 below. In particular, the closure $\overline{\nabla f(\mathbb{R}^n)}$ of the gradient space $\nabla f(\mathbb{R}^n)$ is convex, and both $\nabla f(x(t))$ and $\nabla f(x_i)$ converge to the point p^* in $\overline{\nabla f(\mathbb{R}^n)}$ having the minimum norm, which is uniquely determined.

Theorem 3.14. *Let p^* denote the minimum-norm point of $\overline{\nabla f(\mathbb{R}^n)}$.*

(1) $\nabla f(x(t))$ converges to p^* , and $x(t)/t$ converges to $-p^*$.

(2) $\nabla f(x_i)$ converges to p^* , and x_i/i converges to $-p^*/L$.

Proof. It suffices to show the claims for $x(t)/t$ and x_i/i . The unique minimizer of f^* over the unit sphere is written as $-p^*/\|p^*\|$. Indeed, consider the unit normal vector u^* of supporting hyperplane at p^* , which is written as $u^* = -p^*/\|p^*\|$. Then $f^\infty(v) \geq \langle v, p^* \rangle \geq -\|p^*\| = \langle u^*, p^* \rangle = f^\infty(u^*)$. $\square$

Since $-p^* = \kappa^* \xi^*$, the expected convergences in Questions 3.6 and 3.11 hold in this case. We end this section with other interesting aspects.

Hessian Riemannian gradient flow. Here we point out that the convergence of $\nabla f(x(t))$ to the minimum-norm point p^* can also be explained via the theory of *Hessian Riemannian gradient flow* by Alvarez, Bolte, and Brahic [2]. Suppose for simplicity that Hessian $\nabla^2 f(x)$ is nonsingular for every $x \in \mathbb{R}^n$. Consider the continuous gradient flow $x(t)$, and let $p(t) := \nabla f(x(t))$. One more differentiation in (3.1) yields

$$\dot{p}(t) = -\nabla^2 f(x(t))p(t).$$

From $\nabla^2 f(x(t)) = (\nabla^2 f^*(p(t)))^{-1}$, we have the following differential equation obeyed by $p(t)$:

$$\dot{p}(t) = -(\nabla^2 f^*(p(t)))^{-1}p(t), \quad p(0) = \nabla f(x_0). \quad (3.21)$$

In fact, this can also be viewed as a gradient flow on a Riemannian manifold. Define a Riemannian metric $\langle \cdot, \cdot \rangle^f$ on an open convex set $\nabla f(\mathbb{R}^n)$ by

$$\langle u, v \rangle^f := \langle u, \nabla^2 f^*(p)v \rangle \quad (u, v \in T_p = \mathbb{R}^n, p \in \nabla f(\mathbb{R}^n)). \quad (3.22)$$

In this metric, the gradient $\nabla^f h(p)$ of $h : \nabla f(\mathbb{R}^n) \rightarrow \mathbb{R}$ is given by $(\nabla^2 f^*(p))^{-1} \nabla h(p)$. Then the differential equation (3.21) is interpreted as the gradient flow of the squared-norm function $p \mapsto \|p\|^2/2$:

$$\dot{p}(t) = -\nabla^f \frac{\|p(t)\|^2}{2}, \quad p(0) = \nabla f(x_0). \quad (3.23)$$

This is a particular instance of a Hessian Riemannian gradient flow in [2]. Then, by [2, Proposition 4.4], the solution $p(t)$ of (3.23) minimizes $\|p\|^2/2$ over $\overline{\nabla f(\mathbb{R}^n)}$ in limit $t \rightarrow \infty$. which proves $\lim_{t \rightarrow \infty} \nabla f(p(t)) = p^*$, the first part of Theorem 3.14 (1).

Mirror descent. The discrete version (Theorem 3.14 (2)) may be explained from the framework of *mirror descent* [42], where we consult [10, Chapter 4] for it. Consider a general optimization problem

$$\text{Min. } g(p) \quad \text{s.t. } p \in \mathcal{D}, \quad (3.24)$$

where g is a differential convex function on an open convex set $\mathcal{D} \subseteq \mathbb{R}^n$. A *mirror map* $\Phi : \mathcal{D} \rightarrow \mathbb{R}$ is a differentiable strictly convex function such that $\nabla \Phi : \mathcal{D} \rightarrow \mathbb{R}^n$ is bijective and $\|\nabla \Phi(p)\| \rightarrow \infty$ if p goes to the boundary of $\mathcal{D}$. The basic form of the *mirror descent algorithm* produces the sequence $p_1, p_2, \dots$, according to the update

$$\nabla \Phi(p_{i+1}) := \nabla \Phi(p_i) - \beta_i \nabla g(p_i), \quad (3.25)$$

where β_i is a step size. Under several assumptions on g, Φ , the average solution $(1/i) \sum_{j=1}^i p_j$ (or the best solution ever) is shown to converge to a minimizer of g ; see e.g., [10, Theorem 4.2]. Also see [6, Section 9.2].

Now, consider the setting $g(p) := \|p\|^2/2$ and $\mathcal{D} := \nabla f(\mathbb{R}^n)$. That is, (3.24) is the minimum-norm point problem on $\nabla f(\mathbb{R}^n)$. As a mirror map, we can choose Legendre-Fenchel conjugate $\Phi := f^*$. Then the update (3.25) becomes

$$\nabla f^*(p_{i+1}) := \nabla f^*(p_i) - \beta_i p_i, \quad (3.26)$$

Define $x_i \in \mathbb{R}^n$ by $x_i := \nabla f^*(p_i)$. Since $p_i = \nabla f(x_i)$, (3.26) becomes

$$x_{i+1} := x_i - \beta_i \nabla f(x_i). \quad (3.27)$$

This is nothing but the gradient descent. Further, the above Hessian Riemannian gradient flow is viewed as the continuous limit. Therefore, Theorem 3.14 (2) is expected from convergence of the mirror descent, although we could not find results on mirror descent that can derive it.

It may be interesting to develop a manifold analogy of these observations, which may use the space $\nabla^\infty f(M) \subseteq CM^\infty$ in [29]. Related to this issue, in Section 4.1, we will consider an analogous gradient flow (Kirwan's flow) in the complex projective space.

Matrix scaling and geometric programming. The *matrix scaling problem* [48] is: For a given nonnegative matrix $A = (a_{ij}) \in \mathbb{R}_+^{n \times m}$, find positive diagonal matrices (scaling matrices) X, Y such that $\|(XAY)\mathbf{1} - \mathbf{1}/n\| \approx 0$ and $\|(XAY)^\top \mathbf{1} - \mathbf{1}/m\| \approx 0$. This scalability condition is represented as $\nabla f_A(x, y) \approx 0$ for the following convex function

$$f_A(x, y) = \log \sum_{i,j} a_{ij} e^{x_i + y_j} - \frac{\mathbf{1}^\top x}{n} - \frac{\mathbf{1}^\top y}{m} \quad (x \in \mathbb{R}^n, y \in \mathbb{R}^m). \quad (3.28)$$

Required scaling matrices X, Y are obtained by $X = e^{\text{diag } x}$, $Y = e^{\text{diag } y}$ for x, y having small gradient norm $\|\nabla f_A(x, y)\|$. Particularly, such a point x, y is obtained by minimizing f_A . *Sinkhorn algorithm* [48]—the standard algorithm for matrix scaling—is viewed as alternating minimization of f_A .

For motivating combinatorial optimization application, Hayashi, Hirai, and Sakabe [28] studied asymptotic behavior of the Sinkhorn algorithm in the unscalable case ($\inf f_A(x, y) = -\infty$), and revealed remarkable convergence properties: Both gradient $\nabla f_A(x, y)$ (= row and column marginal of A) and the scaled matrix $e^{\text{diag } x} A e^{\text{diag } y}$ become oscillating between

two accumulation points in limit, which is characterized by a canonical block-triangular form of A , known as (an extended version of) the *DM-decomposition* [17]; see also [41, Section 2.2.3].

Our results reveal analogous asymptotic behavior of the gradient descent for unscalable matrix scaling. The matrix-scaling optimization for (3.28) falls into a more general class of convex optimization, called *geometric programming*, to which our results are applicable. The geometric programming asks to minimize a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ of the following form:

$$f(x) = \log \sum_{\ell=1}^N a_{\ell} e^{\omega_{\ell}^{\top} x} \quad (x \in \mathbb{R}^n), \quad (3.29)$$

where $a_{\ell} > 0$ and $\omega_{\ell} \in \mathbb{R}^n$ for $\ell = 1, 2, \dots, N$. It is well-known (see e.g., [13]) that

- f is L -smooth convex with $L := \max_{\ell} \|\omega_{\ell}\|^2$, and
- $\overline{\nabla f(\mathbb{R}^n)} = \text{Conv}\{\omega_{\ell}\}_{\ell \in [N]}$.

Therefore, with $L = 2$, by the gradient descent (3.9) applied to (3.28), the gradient sequence $\nabla f_A(x_i)$ converges to the minimum-norm point p^* of $\text{Conv}\{e_i + e_j \mid i, j : a_{ij} > 0\}$. The point p^* and the limit of $e^{\text{diag } x_i} A e^{\text{diag } y_i}$ (in the projective space) are again characterized via the DM-decomposition of A . They will be explained for a more general setting of operator scaling in Section 4.2.

4 Applications

4.1 Norm-minimization in reductive group actions

We consider the formulation of *noncommutative optimization* in [12]; see also [30]. Let $G \subseteq GL_n$ be a connected reductive algebraic group over $\mathbb{C}$. Specifically, it is closed under complex conjugate $g \in G \Rightarrow g^{\dagger} \in G$ and is defined by a finite set of polynomials. Its Lie algebra $\mathfrak{g}$ is the complexification of the Lie algebra $\mathfrak{k}$ of a maximal compact subgroup $K = G \cap U_n$ as $\mathfrak{g} = \mathfrak{k} + i\mathfrak{k}$, where $i\mathfrak{k} \subseteq \mathfrak{p}_{\mathfrak{n}}$. The inner product $\langle, \rangle$ on $\mathfrak{g}$ is defined by $\langle X, Y \rangle := \text{Re tr } XY^{\dagger}$. Let V be a finite dimensional vector space over $\mathbb{C}$. Let $\pi : G \rightarrow GL(V)$ be a rational representation, where Π denotes its Lie algebra representation: $\Pi(X) := (d/dt)\pi(e^{tX})|_{t=0}$. Consider a K -invariant Hermitian inner product $(,)$ and the associated norm $\|\cdot\| = \sqrt{(,)}$ on V . The norm-minimization problem over the orbit $\pi(G)v$ of $v \in V \setminus \{0\}$ is given by

$$\inf. \quad \|\pi(g)v\| \quad \text{s.t.} \quad g \in G. \quad (4.1)$$

It turned out (e.g., [12]) that this optimization problem and relatives have numerous, sometimes unexpected, applications and connections in various fields of mathematical sciences. The most fundamental problem is to ask whether the infimum is zero, i.e., the origin 0 is in an orbit closure $\overline{\pi(G)v}$. This is the semi-stability problem in geometric invariant theory (GIT). The representation π gives rise to a Hamiltonian action on complex projective space $\mathbb{P}(V)$ by $(g, [v]) \mapsto [\pi(g)v]$. The corresponding (modified) *moment map* $\tilde{\mu} : V \rightarrow i\mathfrak{k}$ is given by

$$\langle \tilde{\mu}(v), H \rangle := \frac{(v, \Pi(H)v)}{(v, v)} \quad (v \in V, H \in i\mathfrak{k}), \quad (4.2)$$

where the moment map $\mu : \mathbb{P}(V) \rightarrow \mathfrak{k}$ is formally given by $\tilde{\mu}(v) = i\mu([v])$; see [24, Lemma 8.2]. The following theorem is fundamental:

Theorem 4.1 (Kempf-Ness theorem, Hilbert-Mumford criterion; see [24, Theorem 8.5 (i), Theorem 12.4]). *For $v \in V \setminus \{0\}$, the following conditions are equivalent:*

- (i) $\inf_{g \in G} \|\pi(g)v\| = 0$.
- (ii) $\inf_{g \in G} \|\tilde{\mu}(\pi(g)v)\| > 0$.
- (iii) *There is a 1-parameter subgroup $t \mapsto e(t)$ of G such that $\lim_{t \rightarrow \infty} \pi(e(t))v = 0$.*

The orbit $\pi(G)v$ in this situation is called *unstable*. Accordingly, we call the 1-parameter subgroup $e(t)$ in (iii) a *destabilizing 1-PSG*.

The instability corresponds to the lower-unboundedness of the *Kempf-Ness function* $g \mapsto \log \|\pi(g)v\|$. Since $\|\cdot\|$ is K -invariant, the Kempf-Ness function is viewed as a function on the symmetric space $K \backslash G$. With $K \backslash G \simeq P_n \cap G$ by $Kg \mapsto g^\dagger g$ and $\|\pi(g)v\|^2 = (\pi(g^\dagger g)v, v)$, we may consider the Kempf-Ness function as the following function F_v on Hadamard manifold $P_n \cap G$:

$$F_v(x) := \log(\pi(x)v, v) \quad (x \in P_n \cap G). \quad (4.3)$$

Then, F_v is an L -smooth convex function such that the transported gradient of F_v provides the moment map $\tilde{\mu}$:

Lemma 4.2 ([12]). (1) F_v is N_π^2 -smooth convex, where N_π is the maximum of the norm of a weight for π .

$$(2) \quad \tau_{x \rightarrow I} \nabla F_v(x) = \tilde{\mu}(\pi(x^{1/2})v).$$

The second property (2) is implicit in [12] and follows from $\tau_{x \rightarrow I} H = x^{-1/2} H x^{-1/2}$ and $\langle \nabla F_v(x), H \rangle_x = (d/dt) F_v(x^{1/2} e^{tx^{-1/2} H x^{-1/2}} x^{1/2})|_{t=0} = \langle \tilde{\mu}(\pi(x^{1/2})v), x^{-1/2} H x^{-1/2} \rangle_I$. In particular, for the Kempf-Ness function F_v , the unboundedness and the positivity of the minimum gradient-norm are equivalent. Applying Corollary 3.3, we have:

Theorem 4.3. $\inf_{g \in G} \|\tilde{\mu}(\pi(g)v)\| = \sup_{\xi \in B_I} -F_v^\infty(\xi)$. If $F_v^\infty(\xi) < 0$, then $t \mapsto e^{t\xi}$ is a destabilizing 1-PSG.

Proof. $\inf_{g \in G} \|\tilde{\mu}(\pi(g)v)\| = \inf_{x \in P_n \cap G} \|\tilde{\mu}(\pi(x^{1/2})v)\|$ follows from $\tilde{\mu}(\pi(ug)v) = u\tilde{\mu}(\pi(g)v)u^\dagger$ for $u \in K$, the polar decomposition $g = ux$ for $u \in K$, $x \in P_n \cap G$, and $x \in P_n \cap G \Rightarrow x^a \in P_n \cap G$ (since G is algebraic). The latter part can be seen from the definitions of the Kempf-Ness function (4.3) and the recession function (2.1). $\square$

As seen below, this is a part of the theory of moment-weight inequality [24], in which the recession function F_v^∞ is essentially Mumford's numerical invariant, called the μ -weight; see Lemma 4.11 below.

Bürgisser et al. [12] applies the gradient descent (the *first-order algorithm*) to F_v :

$$x_{k+1} = \exp_{x_k} \left(-\frac{1}{L} \nabla F_v(x_k) \right), \quad x_0 = I, \quad (4.4)$$

where $L := N_\pi^2$. For the semi-stable case ($\text{cap}(v) := \inf_{g \in G} \|\pi(g)v\| > 0$), they showed its iteration complexity to compute $\text{cap}(v)$ and to find $g \in G$ with $\|\tilde{\mu}(\pi(g)v)\| \approx 0$. For unstable case ($\text{cap}(v) = 0$), our result (Theorem 3.7) implies that the gradient descent constructs a destabilizing 1-PSG in limit, where a destabilizing 1-PSG is said to be *maximum* if it is obtained as the minimizer of F_v^∞ over S_I .

Theorem 4.4. *Suppose that $\inf_{g \in G} \|\pi(g)v\| = 0$. Let x_k be the sequence of (4.4), and let u_k be the sequence defined by $x_k = e^{d(x_k, I)u_k}$. Then u_k converges to the unique minimizer ξ^* of F_v^∞ over S_I , where $t \mapsto e^{t\xi^*}$ is the maximum destabilizing 1-PSG.*

Unfortunately, since F_v^∞ is not necessarily (upper-semi)continuous, this theorem does not imply the algorithmic statement: $t \mapsto e^{tu_k}$ is a destabilizing 1-PSG for some large k . Therefore, we need a certain rounding idea to obtain a destabilizing 1-PSG from u_k . We see in the next Section 4.2 that such a rounding is possible for the left-right action.

We also consider the convergence of the moment map (transported gradient). Let $\mathcal{C}_\pi \subseteq i\mathfrak{k}$ denote a positive Weyl chamber: It is a convex cone with the property that for any $H \in i\mathfrak{k}$ there is a unique point $\text{spec } H \in \mathcal{C}_\pi$ with $\text{spec } H = kHk^\dagger$ for some $k \in K$. The *moment polytope* $\Delta_v \subseteq \mathcal{C}_\pi$ is defined as the closure of the image of $g \mapsto \text{spec } \tilde{\mu}(\pi(g)v)$:

$$\Delta_v := \overline{\{\text{spec } \tilde{\mu}(\pi(g)v) \mid g \in G\}}.$$

The convexity theorem by Guillemin and Sternberg [25] says that it is a convex polytope.

Theorem 4.5 (Convexity theorem [25]). *Δ_v is a convex polytope.*

From Theorems 3.1 and 3.7 and

$$\inf_{x \in P_n \cap G} \|\nabla F_v(x)\| = \inf_{g \in G} \|\tilde{\mu}(\pi(g)v)\| = \inf_{g \in G} \|\text{spec } \tilde{\mu}(\pi(g)v)\| = \inf_{p \in \Delta_v} \|p\| \quad (4.5)$$

we have the convergence of the moment-map spectre along the gradient-descent trajectory:

Theorem 4.6. *$\text{spec } \tilde{\mu}(\pi(x_k^{1/2})v)$ converges to the minimum norm point p^* of Δ_v .*

Remark 4.7. Suppose that the convergence in Question 3.11 is true. Then $\tilde{\mu}(\pi(x_k^{1/2})v) = \tau_{x_k \rightarrow I} \nabla F_v(x_k)$ converges to $-\kappa^* \xi^*$, where $\kappa^* = \inf_{g \in G} \|\tilde{\mu}(\pi(g)v)\| = \inf_{x \in P_n \cap G} \|\nabla F_v(x)\|$ and ξ^* is the unique minimizer of F_v^∞ over S_I . On the other hand, $\kappa^* \xi^*$ is the limit of the sequence H_k determined by $x_k = e^{kH_k/L}$ by Theorem 3.7 and Proposition 3.9 (3). Choose $\sigma_k \in K$ so that $\text{spec } H_k = \sigma_k H_k \sigma_k^\dagger$. Then the analogue of Theorem 3.14 holds:

$$\lim_{k \rightarrow \infty} \sigma_k^\dagger \tilde{\mu}(\pi(x_k^{1/2})v) \sigma_k = p^*, \quad \lim_{k \rightarrow \infty} \text{spec } H_k = -p^*.$$

We see in Theorem 4.6 below that it is true for the continuous setting.

Moment-weight inequality and gradient flow of moment-map squared. Clearly, via Theorem 3.1, the above results (Theorems 4.4 and 4.6) hold for the continuous-time gradient flow:

$$\dot{x}(t) = -\nabla F_v(x(t)), \quad x(0) = I, \quad (4.6)$$

Our consideration for this case falls into the theory of *moment-weight inequality* by Georgoulas, Robbin, and Salamon [24], which builds upon the earlier work of GIT by Kempf, Kirwan, Mumford, Ness, and a recent work by Chen and Sun [15] on K -stability. Here we briefly summarize the relation by deducing an important part of the theory from our results in Section 3.1. We use notation $g \cdot [v] := [\pi(g)v]$ for the action on $\mathbb{P}(V)$. According to [24, Chapter 3], consider the gradient flow (*Kirwan's flow*) of the squared-norm of the moment-map on $\mathbb{P}(V)$:

$$\dot{\zeta}(t) = -\nabla \frac{\|\mu(\zeta(t))\|^2}{2}, \quad \zeta(0) = [v]. \quad (4.7)$$

This is a gradient flow of a real analytic function $\zeta \mapsto \|\mu(\zeta)\|^2/2$ on a compact manifold $\mathbb{P}(V)$. By a standard argument of Lojasiewics gradient inequality, the limit of $\zeta(t)$ exists.

Theorem 4.8 (Convergence Theorem [24, Theorem 3.3]). *The limit $\zeta_\infty := \lim_{t \rightarrow \infty} \zeta(t)$ exists.*

Further, the limit ζ_∞ attains the infimum of the moment-map norm over the orbit $G \cdot [v]$ in $\mathbb{P}(V)$.

Theorem 4.9 (Moment-limit theorem [24, Theorem 6.4]). *Let $\zeta(t)$ be the solution of (4.7), and let $\zeta_\infty := \lim_{t \rightarrow \infty} \zeta(t)$. Then it holds*

$$\|\mu(\zeta_\infty)\| = \inf_{g \in G} \|\mu(g \cdot [v])\|. \quad (4.8)$$

The equality (4.8) also follows from Theorem 3.1, since the solution ζ is obtained from $x(t)$ as follows:

Proposition 4.10. *The solution $\zeta(t)$ of (4.7) is represented as $\zeta(t) = x(2t)^{1/2} \cdot [v]$ for the solution $x(t)$ of (4.6).*

This is a variant of [24, Theorem 4.1 (ii)], and the proof is also similar; we give it in the Appendix. In particular, Theorem 4.6 can be understood as a discrete version of the moment-limit theorem, though we do not know whether $\zeta_k := x_k^{1/2} \cdot [v]$ converges.

We next explain the moment-weight inequality. The (restriction of) μ -weight $\tilde{w}_\mu : \mathbb{P}(V) \times i\mathfrak{k} \rightarrow \mathbb{R}$ is defined by

$$\tilde{w}_\mu([v], H) := \lim_{t \rightarrow \infty} \operatorname{tr} \tilde{\mu}(\pi(e^{tH})v)H \quad ([v] \in \mathbb{P}(V), H \in i\mathfrak{k} = T_I(P_n \cap G)). \quad (4.9)$$

The μ -weight is nothing but the recession function of F_v^∞ .

Lemma 4.11 (See [24, Lemma 5.2]). $\tilde{w}_\mu([v], H) = F_v^\infty(H)$.

Proof. $2F_v^\infty(H) = \lim_{t \rightarrow \infty} F_v(e^{2tH})/t = \lim_{t \rightarrow \infty} (d/dt)F_v(e^{2tH}) = 2 \lim_{t \rightarrow \infty} \operatorname{tr} \tilde{\mu}(\pi(e^{tH})v)H = 2\tilde{w}_\mu([v], H)$. $\square$

We now state a main part of moment-weight inequality (for linear actions).

Theorem 4.12 (Moment-weight inequality [24, Theorems 6.7, 10.1, 10.2, 10.4]). *It holds*

$$\inf_{g \in G} \|\mu(g \cdot [v])\| \geq \sup_{H \in i\mathfrak{k} \setminus \{0\}} \frac{-\tilde{w}_\mu([v], H)}{\|H\|}. \quad (4.10)$$

Suppose that $\kappa^ := \inf_{g \in G} \|\mu(g \cdot [v])\| > 0$. Then the equality in (4.10) holds, and the supremum is attained by unique $H^* \in i\mathfrak{k}$ with $\|H^*\| = 1$, obtained as follows: Let $H(t)$ be defined via $x(t) = e^{tH(t)}$ for solution $x(t)$ of (4.6). Then the limit $H_\infty := \lim_{t \rightarrow \infty} H(t)$ exists, $\|H_\infty\| = \kappa^*$, and $H^* = H_\infty/\|H_\infty\|$.*

From our convex-optimization perspective, the moment-weight inequality (4.10) is explained by the weak duality (Lemma 2.2). The equality case is explained by the strong duality (Theorem 3.1), the gradient-flow construction of the unique minimizer of F_v^∞ , and the formula of the velocity of escape (Proposition 3.4).

We also describe the limit of the moment map $\tilde{\mu}$, which is a sharpening of [24, (10.9) in Theorem 10.4], and an analogue of Theorem 3.14 (1).

Theorem 4.13 (Limit of moment map). *Suppose that $\kappa^* := \inf_{g \in G} \|\mu(g \cdot [v])\| > 0$. It holds $\lim_{t \rightarrow \infty} \tilde{\mu}(\pi(x^{1/2}(t))v) = -H_\infty (= i\mu(\zeta_\infty))$. In particular, for $H(t) = \sigma(t) \operatorname{spec} H(t) \sigma(t)^\dagger$, it holds*

$$\lim_{t \rightarrow \infty} \sigma(t)^\dagger \tilde{\mu}(\pi(x^{1/2}(t))v) \sigma(t) = p^*, \quad \lim_{t \rightarrow \infty} \operatorname{spec} H(t) = -p^*.$$

Proof. By the convergence theorem (Theorem 4.8) and Proposition 4.10, $\tilde{\mu}(\pi(x^{1/2}(t))v) = i\mu(x^{1/2} \cdot [v])$ converges to $i\mu(\zeta_\infty)$. By Lemma 4.2 (2) and Proposition 3.5, it must hold $i\mu(\zeta_\infty) = -\kappa^* H^* = -H_\infty = -\lim_{t \rightarrow \infty} H(t)$. $\square$

In particular, the convergence in Question 3.6 holds in this case. We finally state one important uniqueness property of minimizers of the moment-map norm over $\pi(\overline{G \cdot [v]})$, which seems not follow from our approach directly.

Theorem 4.14 (Second Ness uniqueness theorem [24, Theorem 6.5]). *For $\zeta, \zeta' \in \overline{G \cdot [v]}$, if $\|\mu(\zeta)\| = \|\mu(\zeta')\| = \inf_{g \in G} \|\mu(g \cdot [v])\|$, then $\zeta' \in K \cdot \zeta$.*

In the next subsection, we characterize such minimizers in the left-right action.

4.2 Operator scaling and its gradient-flow limit

Let $A = (A_1, A_2, \dots, A_N) \in \mathbb{C}^{N(n \times m)}$ be an N -tuple of $n \times m$ matrix over $\mathbb{C}$. Let $p \in \mathbb{R}_+^n$, $q \in \mathbb{R}_+^m$ be nonnegative vectors with the same sum $\sum_i p_i = \sum_j q_j$, where p, q are arranged as

$$p_1 \geq p_2 \geq \dots \geq p_n, \quad q_1 \leq q_2 \leq \dots \leq q_m. \quad (4.11)$$

The *operator scaling* problem, originally introduced by Gurvits [26] for $p = q = \mathbf{1}$ and extended by Franks [18] for general p, q , is to ask: For a given accuracy $\epsilon \geq 0$, find $g \in GL_n, h \in GL_m$ such that

$$\left\| \sum_{\ell=1}^N g A_\ell h^\dagger h A_\ell^\dagger g^\dagger - \operatorname{diag} p \right\|^2 + \left\| \sum_{\ell=1}^N h A_\ell^\dagger g^\dagger g A_\ell h^\dagger - \operatorname{diag} q \right\|^2 \leq \epsilon^2, \quad (4.12)$$

where the norm is the Frobenius norm. A matrix tuple A is said to be (approximately) (p, q) -scalable if for every positive $\epsilon > 0$ there are $g \in GL_n, h \in GL_m$ satisfying (4.12). If some g, h satisfy (4.12) for $\epsilon = 0$, then A is called *exactly* (p, q) -scalable, and $gAh^\dagger$ is called a (p, q) -scaling of A . The operator scaling is a quantum generalization of the matrix scaling, and turned out to have rich applications; see [18, 21, 22, 23]. For simplicity, we assume that the left and right common kernels of A are both trivial: $\bigcap_\ell \ker A_\ell = \{0\}$ and $\bigcap_\ell \ker A_\ell^\dagger = \{0\}$.

In the view of the previous section, the operator scaling problem can be viewed as the moment polytope membership of the *left-right action* $\pi : SL_n \times SL_m \rightarrow GL(\mathbb{C}^{N(n \times m)})$ defined by

$$\pi(g, h)(B) := gBh^\dagger = (gB_1h^\dagger, gB_2h^\dagger, \dots, gB_Nh^\dagger), \quad (4.13)$$

where $B = (B_1, B_2, \dots, B_N) \in \mathbb{C}^{N(n \times m)}$. A maximal compact subgroup K is given as $SU_n \times SU_m$, and the K -invariant Hermitian product on $V = \mathbb{C}^{N(n \times m)}$ is given by $\langle B, C \rangle := \sum_{\ell=1}^N \operatorname{tr} B_\ell C_\ell^\dagger$. From $\Pi(X, Y)(B) = XB + BY^\dagger$, we see that the moment map $\tilde{\mu} : \mathbb{C}^{N(n \times m)} \rightarrow \mathfrak{p}_n^1 \times \mathfrak{p}_m^1$ is given by

$$\tilde{\mu}(B) = \frac{1}{\|B\|^2} \left(\sum_{\ell=1}^N B_\ell B_\ell^\dagger, \sum_{\ell=1}^N B_\ell^\dagger B_\ell \right) - \left(\frac{1}{n} I, \frac{1}{m} I \right). \quad (4.14)$$

A positive Weyl chamber is taken as the set of diagonal matrices $(\text{diag } p, \text{diag } q)$ with p, q satisfying (4.11) and $\mathbf{1}^\top p = \mathbf{1}^\top q = 0$. We regarded it as a subset of $\mathbb{R}^n \times \mathbb{R}^m$. Then the moment polytope Δ_A consists of vectors of eigenvalues of $\tilde{\mu}(B)$ over $B \in \overline{SL_n \cdot A \cdot SL_m}$ (the closure of the $SL_n \times SL_m$ -orbit of A). Comparing (4.14) with (4.12), we have:

Lemma 4.15. *A is (p, q) -scalable if and only if $(p/c - \mathbf{1}/n, q/c - \mathbf{1}/m)$ belongs to Δ_A , where $c := \sum_i p_i = \sum_j q_j$.*

We consider the operator scaling problem of the most basic case: $(p, q) = (\mathbf{1}/n, \mathbf{1}/m)$. Then, it holds

$$A \text{ is } (\mathbf{1}/n, \mathbf{1}/m)\text{-scalable} \iff (0, 0) \in \Delta_A \iff \inf_{g, h} \|\tilde{\mu}(gAh^\dagger)\| = 0. \quad (4.15)$$

Accordingly, Kempf-Ness theorem with Hilbert-Mumford criterion (Theorem 4.1) links with the $(\mathbf{1}/n, \mathbf{1}/m)$ -scaling problem, and is sharpened as follows: Let $\mathcal{S}_A$ denote the family of pairs of vector subspaces $X \subseteq \mathbb{C}^n$, $Y \subseteq \mathbb{C}^m$ such that $u^\top A_\ell \bar{v} = 0$ for all $u \in X$, $v \in Y$, $\ell \in [N]$.

Theorem 4.16 (Charaterization of scalability [26]). *The following are equivalent:*

- (i) $\inf_{g \in SL_n, h \in SL_m} \|gAh^\dagger\| > 0$.
- (ii) *A is $(\mathbf{1}/n, \mathbf{1}/m)$ -scalable.*
- (iii) *For all $(X, Y) \in \mathcal{S}_A$, it holds $(1/n) \dim X + (1/m) \dim Y \leq 1$.*

A vector-space pair $(X, Y) \in \mathcal{S}_A$ violating (iii) actually gives rise to a destabilizing 1-PSG as follows: Choose $\sigma \in SU_n$ and $\tau \in SU_m$ such that the first r rows of σ span X and the first s rows of τ span Y , where $(r, s) := (\dim X, \dim Y)$. Then one can see that $t \mapsto (e^{t \text{diag}(\mathbf{1}_{[r]} - (r/n)\mathbf{1})} \sigma, e^{t \text{diag}(\mathbf{1}_{[s]} - (s/m)\mathbf{1})} \tau)$ is a destabilizing 1-PSG.

Further, the strictly inequality in (iii) brings the exact scalability:

Theorem 4.17 (Exact scalability [26]). *If $(1/n) \dim X + (1/m) \dim Y < 1$ for all $(X, Y) \in \mathcal{S}_A$ other than $(\{0\}, \mathbb{C}^m)$ and $(\mathbb{C}^n, \{0\})$, then A is exactly $(\mathbf{1}/n, \mathbf{1}/m)$ -scalable.*

The exact case corresponds to the existence of g, h with $\tilde{\mu}(gAh^\dagger) = 0$. By Lemma 4.2 (2), this is the case where the Kempf-Ness function F_A has an optimum = a point of zero gradient. Then, Theorem 4.17 can be deduced from a general property (2.2) of the recession function F_A^∞ (given in (4.18) below). Here the Kempf-Ness function $F_A : P_n^1 \times P_m^1 \rightarrow \mathbb{R}$ is written as

$$F_A(x, y) := \log \text{tr} \sum_{\ell=1}^N x A_\ell y A_\ell^\dagger \quad (x \in P_n^1, y \in P_m^1). \quad (4.16)$$

Lemma 4.18 ([12]). *F_A is 2-smooth convex.*

Now Theorem 4.4 (Corollary 3.3, or moment-weight inequality (Theorem 4.12)) sharpens (ii) $\Leftrightarrow$ (iii) of Theorem 4.16 in the following min-max (inf-sup) form:

Theorem 4.19 (Duality theorem for scalability limit).

$$\begin{aligned} & \inf_{g, h} \left\| \left(\sum_{\ell=1}^N g A_\ell h^\dagger h A_\ell^\dagger g^\dagger - \frac{1}{n} I, \sum_{\ell=1}^N h A_\ell^\dagger g^\dagger g A_\ell h - \frac{1}{m} I \right) \right\| \\ &= \sup_{p, q, \sigma, \tau} -\max\{p_i + q_j \mid \exists \ell, (\sigma A_\ell \tau^\dagger)_{ij} \neq 0\}, \end{aligned} \quad (4.17)$$

where the infimum in LHS is taken over all $g \in GL_n$, $h \in GL_m$ with $\|gAh^\dagger\| = 1$ and the supremum in RHS is taken over all $\sigma \in SU_n$, $\tau \in SU_m$, $p \in \mathbb{R}^n$, $q \in \mathbb{R}^m$ with $\|(p, q)\| \leq 1$ and $\mathbf{1}^\top p = \mathbf{1}^\top q = 0$.

Proof. It suffices to show that $-F_A^\infty$ is equal to the objective function of RHS in (4.17). Here $(H, G) \in \mathfrak{p}_n^1 \times \mathfrak{p}_m^1$ is written as $(H, G) = (\sigma^\dagger \text{diag } p\sigma, \tau^\dagger \text{diag } q\tau)$ for $\sigma \in SU_n$, $\tau \in SU_m$, $p \in \mathbb{R}^n$, $q \in \mathbb{R}^m$ with $\mathbf{1}^\top p = \mathbf{1}^\top q = 0$. Then we have

$$\begin{aligned} F_A^\infty(H, G) &= \lim_{t \rightarrow \infty} \frac{1}{t} \log \text{tr} \sum_{\ell} e^{tH} A_{\ell} e^{tG} A_{\ell}^\dagger = \lim_{t \rightarrow \infty} \frac{1}{t} \log \sum_{\ell, i, j} |(\sigma A_{\ell} \tau^\dagger)_{ij}|^2 e^{t(p_i + q_j)} \\ &= \max\{p_i + q_j \mid \exists \ell, (\sigma A_{\ell} \tau^\dagger)_{ij} \neq 0\}, \end{aligned} \quad (4.18)$$

where we used $\lim_{t \rightarrow \infty} \frac{1}{t} \log \sum_k e^{c_k + td_k} = \max_k d_k$ in the last equality. $\square$

Consider the continuous-time and discrete-time gradient flows of F_A :

$$(\dot{x}(t), \dot{y}(t)) = -\nabla F_A(x(t), y(t)), \quad (x(0), y(0)) = (I, I), \quad (4.19)$$

$$(x_{k+1}, y_{k+1}) = \exp_{x_k, y_k} \left(-\frac{1}{L} \nabla F_A(x_k, y_k) \right), \quad (x_0, y_0) = (I, I), \quad (4.20)$$

where we let $L := 2$ by Lemma 4.18. We study asymptotic behavior of (4.19) and (4.20). In the rest of this section, we are going to characterize:

- (A) The limit of $\text{spec } \tilde{\mu}(x_k^{1/2} A y_k^{1/2}) \Rightarrow$ the minimum-norm point of Δ_A .
- (B) The limit of (x_k, y_k) in cone topology $\Rightarrow$ the unique minimizer of F_A^∞ .
- (C) The limit of $[x_k^{1/2} A y_k^{1/2}]$ in $\mathbb{P}(\mathbb{C}^{N(n \times m)}) \Rightarrow$ the minimizer of moment-map norm $\|\mu\|$ over $[SL_n \cdot A \cdot SL_m]$ (modulo $SU_n \times SU_m$ action).

We show that these are characterized by a certain simultaneous block-triangular form of A . This block-triangular form is a vector-space generalization of the classical *Dulmage-Mendelsohn decomposition* [17] (*DM-decomposition* for short) for a bipartite graph and its associated matrix. We introduce our generalized DM-decomposition in a way analogous to [28, Section 3] for the classical setting, where the essential idea of the construction can be partly found in Ito, Iwata, and Murota [32]. Iwamasa, Oki, and Soma [35] pointed out that our DM-decomposition is a special case of the *Harder-Narasimhan filtration* for a generalized Kronecker quiver.

Recall the family $\mathcal{S}_A$ defined before Theorem 4.16. Define a map $\phi : \mathcal{S}_A \rightarrow \mathbb{R}_+^2$ by

$$\phi(X, Y) := (\dim X, \dim Y) \quad ((X, Y) \in \mathcal{S}_A).$$

Consider the convex hull $\text{Conv } \phi(\mathcal{S}_A) \subseteq \mathbb{R}_+^2$; see the left of Figure 1. Let $\mathcal{E}_A$ denote the subset of $(X, Y) \in \mathcal{S}_A$ such that $\phi(X, Y)$ is an extreme point of $\text{Conv } \phi(\mathcal{S}_A)$ not equal to $(0, 0)$.

Lemma 4.20. *For $(X, Y), (X', Y') \in \mathcal{E}_A$, if $\dim X \leq \dim X'$ and $\dim Y \geq \dim Y'$, then $X \subseteq X'$ and $Y \supseteq Y'$. In particular, ϕ is injective on $\mathcal{E}_A$.*

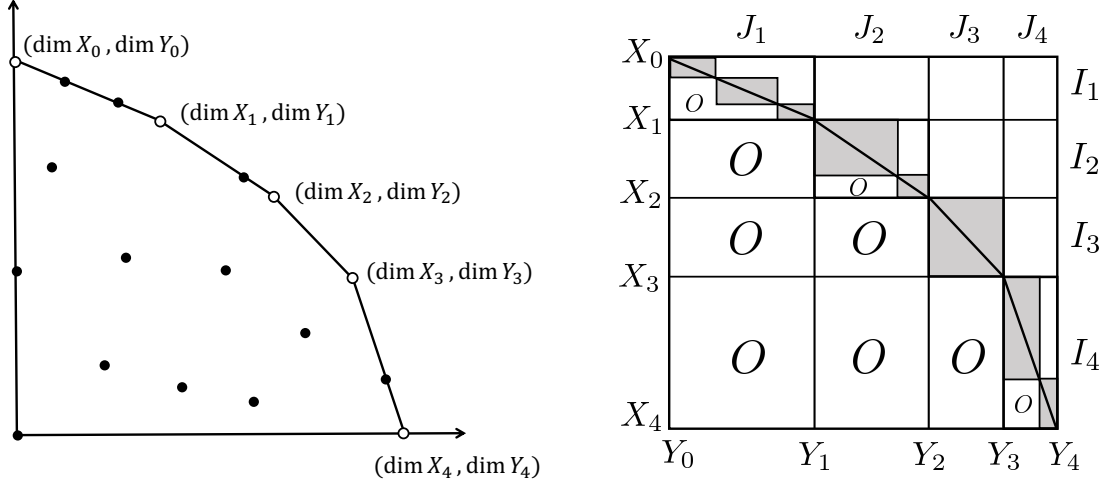


Figure 1: $\text{Conv } \phi(\mathcal{S}_A)$ in (y, x) -plane (left) and a DM-decomposition of A (right)

Proof. We may suppose that $\phi(X, Y)$ and $\phi(X', Y')$ are equal or on an adjacent pair of extreme points. Observe $(X \cap X', Y + Y'), (X + X', Y \cap Y') \in \mathcal{S}_A$. By the dimension identity of vector spaces, it holds

$$\phi(X \cap X', Y + Y') + \phi(X + X', Y \cap Y') = \phi(X, Y) + \phi(X', Y'). \quad (4.21)$$

This implies that $X' = X + X'$ and $Y' = Y \cap Y'$. Otherwise, $\phi(X \cap X', Y + Y')$ or $\phi(X + X', Y \cap Y')$ goes beyond $\text{Conv } \phi(\mathcal{S}_A)$, which contradicts $(X \cap X', Y + Y'), (X + X', Y \cap Y') \in \mathcal{S}_A$. $\square$

Therefore, $\mathcal{E}_A = \{(X_\alpha, Y_\alpha)\}_{\alpha=0}^\theta$ can be arranged as

$$\begin{aligned} \mathbb{C}^n &= X_0 \supset X_1 \supset \cdots \supset X_\theta = \{0\}, \\ \{0\} &= Y_0 \subset Y_1 \subset \cdots \subset Y_\theta = \mathbb{C}^m. \end{aligned} \quad (4.22)$$

For each $\alpha \in [\theta]$, let $\mathcal{L}_A^\alpha$ denote the subset consisting of $(X, Y) \in \mathcal{S}_A$ such that $\phi(X, Y)$ belongs to the edge between $\phi(X_{\alpha-1}, Y_{\alpha-1})$ and $\phi(X_\alpha, Y_\alpha)$.

Lemma 4.21. *If $(X, Y), (X', Y') \in \mathcal{L}_A^\alpha$, then $(X + X', Y \cap Y'), (X \cap X', Y + Y') \in \mathcal{L}_A^\alpha$. In particular, $\mathcal{L}_A^\alpha$ is a modular lattice with respect to the partial order $(X, Y) \preceq (X', Y') \Leftrightarrow X \supseteq X', Y \subseteq Y'$, where the minimum and maximum elements are given by $(X_{\alpha-1}, Y_{\alpha-1})$ and (X_α, Y_α) , respectively.*

For each $\alpha \in [\theta]$, consider a maximal chain (flag) of $\mathcal{L}_A^\alpha$:

$$\begin{aligned} X_{\alpha-1} &= X_{\alpha,0} \supset X_{\alpha,1} \supset \cdots \supset X_{\alpha,\theta_\alpha} = X_\alpha, \\ Y_{\alpha-1} &= Y_{\alpha,0} \subset Y_{\alpha,1} \subset \cdots \subset Y_{\alpha,\theta_\alpha} = Y_\alpha, \end{aligned}$$

where the length θ_α of the chain is uniquely determined by Jordan-Dedekind chain condition. The union $\bigcup_{\alpha=1}^\theta \bigcup_{\beta=0}^{\theta_\alpha} \{X_{\alpha,\beta}\}$ is a maximal chain of the whole lattice $\mathcal{L}_A := \bigcup_{\alpha=1}^\theta \mathcal{L}_A^\alpha$, and is called a *DM-flag*. Its subset $\mathcal{E}_A$ is called the *coarse DM-flag*, which is uniquely determined by A . From a DM-flag, we obtain a simultaneous block triangular form of A as follows. Consider $g \in GL_n$ including, as row vectors, a basis of $X_{\alpha,\beta}$ for each α, β .

Similarly, consider $h \in GL_m$ including, as row vectors, a basis of $Y_{\alpha,\beta}$ for each α, β . Suppose that they are positioned in the latter rows for g and first rows for h . Then, the matrices $B_\ell = gA_\ell h^\dagger$ are simultaneously block-triangularized, as in the right of Figure 1. We call $B = (B_\ell)$ a *DM-decomposition*⁴ of A . When g (resp. h) is restricted to span only X_α (resp. Y_α), it is called a *coarse DM-decomposition* of A .

For abuse of notation, X_α , $X_{\alpha,\beta}$, Y_α , and $Y_{\alpha,\beta}$ also denote the index sets of the corresponding rows and columns of B . Define ordered partitions (I_α) of $[n]$, (J_α) of $[m]$ and their refinements $(I_{\alpha,\beta})$, $(J_{\alpha,\beta})$ by

$$I_\alpha := X_{\alpha-1} \setminus X_\alpha, \quad J_\alpha := Y_\alpha \setminus Y_{\alpha-1} \quad (\alpha \in [\theta]), \quad (4.23)$$

$$I_{\alpha,\beta} := X_{\alpha,\beta-1} \setminus X_{\alpha,\beta}, \quad J_{\alpha,\beta} := Y_{\alpha,\beta} \setminus Y_{\alpha,\beta-1} \quad (\beta \in [\theta_\alpha]). \quad (4.24)$$

Let $\hat{B} = (\hat{B}_\ell)$ denote the matrix tuple of block-diagonal matrices obtained from B_ℓ by replacing each (upper) off-diagonal block $B_k[I_{\alpha,\beta}, J_{\alpha',\beta'}]$ ($(\alpha, \beta) \neq (\alpha', \beta')$) with zero block. We call $\hat{B}$ a *diagonalized DM-decomposition* of A . A diagonalized version of a coarse DM-decomposition is defined analogously.

Let $n_\alpha := |I_\alpha|$ and $m_\alpha := |J_\alpha|$. By convexity of $\text{Conv } \phi(\mathcal{S}_A)$, it holds

$$\frac{n_1}{m_1} < \frac{n_2}{m_2} < \dots < \frac{n_\theta}{m_\theta}. \quad (4.25)$$

Define $(p^*, q^*) \in \mathbb{R}^n \times \mathbb{R}^m$ by

$$p^* := -\frac{1}{n}\mathbf{1} + \frac{1}{C_A} \sum_{\alpha=1}^{\theta} \frac{m_\alpha}{n_\alpha + m_\alpha} \mathbf{1}_{I_\alpha}, \quad q^* := -\frac{1}{m}\mathbf{1} + \frac{1}{C_A} \sum_{\alpha=1}^{\theta} \frac{n_\alpha}{n_\alpha + m_\alpha} \mathbf{1}_{J_\alpha}, \quad (4.26)$$

where the constant C_A is defined by

$$C_A := \sum_{\alpha=1}^{\theta} \frac{n_\alpha m_\alpha}{n_\alpha + m_\alpha}. \quad (4.27)$$

We see from (4.25), (4.26), (4.27) that (p^*, q^*) belongs to the positive Weyl chamber:

$$p_1^* \geq p_2^* \geq \dots \geq p_n^*, \quad q_1^* \leq q_2^* \leq \dots \leq q_m^*, \quad \mathbf{1}^\top p^* = \mathbf{1}^\top q^* = 0. \quad (4.28)$$

Also we observe that

$$p_i^* + q_j^* + \frac{1}{n} + \frac{1}{m} - \frac{1}{C_A} \begin{cases} = 0 & \text{if } (i, j) \in I_\alpha \times J_\alpha, \\ < 0 & \text{if } (i, j) \in I_\alpha \times J_{\alpha'} : \alpha > \alpha', \\ > 0 & \text{if } (i, j) \in I_\alpha \times J_{\alpha'} : \alpha < \alpha'. \end{cases} \quad (4.29)$$

Define $(H^*, G^*) \in S_{I,I}(P_n^1 \times P_m^1)$ by

$$H^* := \frac{-1}{\|(p^*, q^*)\|} (\sigma^*)^\dagger \text{diag } p^* \sigma^*, \quad G^* := \frac{-1}{\|(p^*, q^*)\|} (\tau^*)^\dagger \text{diag } q^* \tau^*, \quad (4.30)$$

where σ^* is a unitary matrix having a basis of X_α in the last n_α rows and τ^* is a unitary matrix having a basis of Y_α in the first m_α rows. By using these notions, we give solutions of (A) and (B):

⁴The classical DM-decomposition restricts $\mathcal{S}_A$ to coordinate subspaces and $\mathcal{L}_A$ to the sublattice of the coordinate subspaces X, Y maximizing $\dim X + \dim Y$, where g, h are chosen as permutation matrices. In this setting, a block-triangular form obtained by using the maximal chain of the whole $\mathcal{L}_A$ was considered by N. Tomizawa (unpublished) in the development of principal partitions in 1970's; see [28, Section 3]. By this reason, our decomposition may be called a *DMT-decomposition*, more precisely.

Theorem 4.22. *Suppose that A is not $(\mathbf{1}/n, \mathbf{1}/m)$ -scalable.*

- (1) (p^*, q^*) is the minimum norm point of Δ_A , and
- (2) (H^*, G^*) is the unique minimizer of F_A^∞ over $S_{I,I}$, where it holds

$$\|(p^*, q^*)\| = -F_A^\infty(H^*, G^*) = \sqrt{\frac{1}{C_A} - \frac{1}{n} - \frac{1}{m}}. \quad (4.31)$$

Proof. We first show (4.31). From the definitions of (p^*, q^*) and C_A , we have

$$\|(p^*, q^*) + (\mathbf{1}/n, \mathbf{1}/m)\|^2 = \frac{1}{C_A^2} \sum_{\alpha=1}^{\theta} \frac{n_\alpha m_\alpha^2}{(n_\alpha + m_\alpha)^2} + \frac{m_\alpha n_\alpha^2}{(n_\alpha + m_\alpha)^2} = \frac{1}{C_A^2} \sum_{\alpha=1}^{\theta} \frac{n_\alpha m_\alpha}{n_\alpha + m_\alpha} = \frac{1}{C_A}.$$

By the last equation in (4.28), we have

$$\|(p^*, q^*)\|^2 = \|(p^*, q^*) + (\mathbf{1}/n, \mathbf{1}/m)\|^2 - \|(\mathbf{1}/n, \mathbf{1}/m)\|^2 = 1/C_A - 1/n - 1/m (> 0).$$

On the other hand, $B = \sigma^* A (\tau^*)^\dagger$ is a coarse DM-decomposition. By (4.29), the maximum $p_i + q_j$ in (4.18) is attained by the index (i, j) of any nonzero element in any diagonal block, and is equal to $-1/n - 1/m + 1/C_A$. Then $F_A^\infty(H^*, G^*) = (1/n + 1/m - 1/C_A) / \|(p, q)\| = -\sqrt{1/C_A - 1/n - 1/m}$.

Thus, to complete the proof, it suffices to show $(p^*, q^*) \in \Delta_A$. $\square$

Proposition 4.23. *Let $\hat{B}$ be a diagonalized DM-decomposition of A .*

- (1) $\hat{B}$ is exactly $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scalable.
- (2) $[\hat{B}] \in \overline{SL_n \cdot A \cdot SL_m}$.

In particular, it holds $(p^, q^*) \in \Delta_A$.*

Proof. (1). We first show:

Claim. $B[I_{\alpha,\beta}, J_{\alpha,\beta}]$ is exactly $(\mathbf{1}/|I_{\alpha,\beta}|, \mathbf{1}/|J_{\alpha,\beta}|)$ -scalable.

Proof of Claim. We can assume that A is already equal to a DM-decomposition B , where all $X_{\alpha,\beta}, Y_{\alpha,\beta}$ are coordinate subspaces. Suppose indirectly that $B[I_{\alpha,\beta}, J_{\alpha,\beta}]$ is not exactly $(\mathbf{1}/|I_{\alpha,\beta}|, \mathbf{1}/|J_{\alpha,\beta}|)$ -scalable. Then, by Theorems 4.16 and 4.17, there is non-trivial $(Z, W) \in \mathcal{S}_{B[I_{\alpha,\beta}, J_{\alpha,\beta}]}$ such that $(1/|I_{\alpha,\beta}|) \dim Z + (1/|J_{\alpha,\beta}|) \dim W \geq 1$. Then $(X_{\alpha,\beta} + Z, Y_{\alpha,\beta-1} + W)$ belongs to $\mathcal{S}_A$. However, $\phi(X_{\alpha,\beta} + Z, Y_{\alpha,\beta-1} + W)$ goes beyond $\text{Conv } \phi(\mathcal{S}_A)$ or lies on the interior of the segment between $\phi(X_{\alpha,\beta-1}, Y_{\alpha,\beta-1})$ and $\phi(X_{\alpha,\beta}, Y_{\alpha,\beta})$. The former case is obviously impossible. The latter case is also impossible due to the maximality of chain $\{(X_{\alpha,\beta}, Y_{\alpha,\beta})\}$ in $\mathcal{L}_A$. $\square$

For each α, β , we can choose scaling matrices $g_{\alpha,\beta}, h_{\alpha,\beta}$ to make $B[I_{\alpha,\beta}, J_{\alpha,\beta}]$ an exact $C_\alpha(\mathbf{1}/|I_{\alpha,\beta}|, \mathbf{1}/|J_{\alpha,\beta}|)$ -scaling, where $C_\alpha := n_\alpha m_\alpha / \{C_A(n_\alpha + m_\alpha)\}$. Then, for $g := \bigoplus_{\alpha,\beta} g_{\alpha,\beta}$, $h := \bigoplus_{\alpha,\beta} h_{\alpha,\beta}$, the scaling $g\hat{B}h^\dagger$ is a desired $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scaling.

(2). Let B be a DM-decomposition of A , where $B \in SL_n \cdot A \cdot SL_m$. For $t > 0$, define $a_t \in SL_n$ and $b_t \in SL_m$ by

$$a_t := e^{t \text{diag } \sum_{\alpha,\beta} (\mathbf{1}_{X_{\alpha,\beta}} - |X_{\alpha,\beta}| \mathbf{1}/n)}, \quad b_t := e^{t \text{diag } \sum_{\alpha,\beta} (\mathbf{1}_{Y_{\alpha,\beta}} - |Y_{\alpha,\beta}| \mathbf{1}/m)}.$$

Then we see

$$a_t B b_t = e^{t \sum_{\alpha, \beta} (1 - |X_{\alpha, \beta}|/n - |Y_{\alpha, \beta}|/m)} (\hat{B} + E_t),$$

where E_t converges to zero for $t \rightarrow \infty$. This implies that $[\hat{B}] \in \overline{[SL_n \cdot A \cdot SL_m]}$. Since $\hat{B}$ admits an exact $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scaling $B^* = g \hat{B} h^\dagger$ and $\|B^*\|^2 = \sum_{\alpha} C_{\alpha} = 1$, we have $\lim_{t \rightarrow \infty} \text{spec } \tilde{\mu}(g a_t B b_t h^\dagger) = \text{spec } \tilde{\mu}(B^*) = (p^*, q^*) \in \Delta_A$. $\square$

Now the sequence of the scaled matrix along the gradient flow/descent accumulates to the $SU_n \times SU_m$ -orbit of a diagonalized DM-decomposition $\hat{B}$, providing a solution of (C):

Theorem 4.24. *Let $\hat{B}$ be a diagonalized DM-decomposition of A , and let B^* be a $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scaling of $\hat{B}$.*

(1) $[x(t)^{1/2} A y(t)^{1/2}]$ converges to a point in $[SU_n \cdot B^* \cdot SU_m]$.

(2) $[x_k^{1/2} A y_k^{1/2}]$ accumulates to points in $[SU_n \cdot B^* \cdot SU_m]$.

Proof. It holds $\tilde{\mu}(B^*) = (\text{diag } p^*, \text{diag } q^*)$. Thus B^* attains the infimum of $\|\tilde{\mu}(B)\|$ over $[B] \in \overline{[SL_n \cdot A \cdot SL_m]}$, which is also the limit of $\|\tilde{\mu}(x(t)^{1/2} A y(t)^{1/2})\|$ and $\|\tilde{\mu}(x_k^{1/2} A y_k^{1/2})\|$. By second Ness uniqueness theorem (Theorem 4.14), we have the claim. $\square$

In particular, $[x(t)^{1/2} A y(t)^{1/2}]$ and $[x_k^{1/2} A y_k^{1/2}]$ converge/accumulate to $\sigma B^* \tau^\dagger$ for a $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scaling B^* , $\sigma \in SU_n$, $\tau \in SU_m$. Although B^* is also a diagonalized DM-decomposition of A , it is not clear how to remove the unitary indeterminacy σ, τ and to extract the DM-structure of B^* . This is possible for the coarse DM-structure as follows:

Theorem 4.25. *Let $(x(t), y(t))$ be the solution of (4.19). Decompose $x(t) = \sigma_t^\dagger e^{\text{diag } p^t} \sigma_t$ and $y(t) = \tau_t^\dagger e^{\text{diag } q^t} \tau_t$, where σ_t and τ_t are unitary matrices, and p^t and q^t are nondecreasing and nonincreasing vectors, respectively.*

(1) $\lim_{t \rightarrow \infty} \frac{(p^t, q^t)}{t} = -(p^*, q^*)$.

(2) $\sigma_t A \tau_t^\dagger$ accumulates to coarse DM-decompositions.

(3) $A^{(t)} := (e^{\text{diag } p^t/2} \sigma_t A \tau_t^\dagger e^{\text{diag } q^t/2}) / \|x(t)^{1/2} A y(t)^{1/2}\|$ accumulates to diagonalized coarse DM-decompositions, which are $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scalings.

Proof. (1). By Theorems 4.12 and 4.13, it holds

$$\lim_{t \rightarrow \infty} \tilde{\mu}(x(t)^{1/2} A y(t)^{1/2}) = -\|(p^*, q^*)\|(H^*, G^*) = -\lim_{t \rightarrow \infty} \frac{(\sigma_t^\dagger \text{diag } p^t \sigma_t, \tau_t^\dagger \text{diag } q^t \tau_t)}{t}.$$

From $\lim_{t \rightarrow \infty} \text{spec } \tilde{\mu}(x_t^{1/2} A y_t^{1/2}) = (p^*, q^*)$, we have (1). Consequently, it holds

$$\lim_{t \rightarrow \infty} \tilde{\mu}(e^{\text{diag } p^t/2} \sigma_t A \tau_t^\dagger e^{\text{diag } q^t/2}) = -\lim_{t \rightarrow \infty} \frac{(\text{diag } p^t, \text{diag } q^t)}{t} = (\text{diag } p^*, \text{diag } q^*).$$

Thus $A^{(t)}$ accumulates to $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scalings.

(3). We are going to show that $A^{(t)}$ becomes block-diagonalized along the coarse DM-structure in limit. We first note

$$\lim_{t \rightarrow \infty} \frac{F_A(x(t), y(t))}{t} = -\lim_{t \rightarrow \infty} \|\nabla F_A(x(t), y(t))\|^2 = -\left(\frac{1}{C_A} - \frac{1}{n} - \frac{1}{m}\right), \quad (4.32)$$

where we used $\lim_{t \rightarrow \infty} F_A(x(t), y(t))/t = \lim_{t \rightarrow \infty} (d/dt)F_A(x(t), y(t))$ for the first equality, and (4.31) with $-F_A^\infty(H^*, G^*) = \lim_{t \rightarrow \infty} \|\nabla F_A(x(t), y(t))\|$ for the second. For any index $(i, j) \in I_\alpha \times J_{\alpha'}$ with $\alpha' > \alpha$, by (1) and (4.29), it holds $-F_A(x(t), y(t)) + p_i^t + q_j^t \rightarrow -\infty$, and hence

$$\lim_{t \rightarrow \infty} (A^{(t)})_{ij} = \lim_{t \rightarrow \infty} e^{(-F_A(x(t), y(t)) + p_i^t + q_j^t)/2} (\sigma_t A \tau_t^\dagger)_{ij} = 0.$$

Namely, all upper triangular blocks of $A^{(t)}$ converge to zero. From the fact that $A^{(t)}$ becomes $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scalings in limit, all lower triangular blocks necessarily converge to zero, implying (3):

$$\lim_{t \rightarrow \infty} (A^{(t)})_{ij} = 0 \quad ((i, j) \in I_\alpha \times J_{\alpha'} : \alpha' < \alpha). \quad (4.33)$$

This convergence can be seen from the following observation on a block-triangular matrix:

$$\begin{pmatrix} A_1 & O \\ B & A_2 \end{pmatrix} \begin{pmatrix} A_1 & O \\ B & A_2 \end{pmatrix}^\dagger = \begin{pmatrix} P_1 & O \\ O & P_2 \end{pmatrix}, \quad \begin{pmatrix} A_1 & O \\ B & A_2 \end{pmatrix}^\dagger \begin{pmatrix} A_1 & O \\ B & A_2 \end{pmatrix} = \begin{pmatrix} Q_1 & O \\ O & Q_2 \end{pmatrix}$$

and $\text{tr } P_1 = \text{tr } Q_1$ imply $B = O$.

(2). By (4.33) and $-F_A(x(t), y(t)) + p_i^t + q_j^t \rightarrow +\infty$ for $(i, j) \in I_\alpha \times J_{\alpha'}$ with $\alpha' < \alpha$, all lower-triangular blocks of $\sigma_t A \tau_t^\dagger$ must converge to zero, implying (2). $\square$

For the discrete setting, we only give a weaker, but still meaningful, result as follows:

Theorem 4.26. *Let (x_k, y_k) be the solution of (4.20). Decompose $x_k = \sigma_k^\dagger e^{\text{diag } p^k} \sigma_k$ and $y_k = \tau_k^\dagger e^{\text{diag } q^k} \tau_k$, where σ_k and τ_k are unitary matrices, and p^k and q^k are nondecreasing and nonincreasing vectors, respectively.*

$$(1) \quad \lim_{k \rightarrow \infty} \frac{L}{k} (p^k, q^k) = -(p^*, q^*).$$

(2) $\sigma_k A \tau_k^\dagger$ accumulates to coarse DM-decompositions. The convergence is linear in the following sense: There are $c > 0$, $M > 0$ such that for all $k \geq M$, $\ell \in [N]$ it holds

$$|(\sigma_k A \tau_k^\dagger)_{\ell j}| \leq e^{-ck} \quad ((i, j) \in I_\alpha \times J_{\alpha'} : \alpha > \alpha').$$

Proof. (1). By Theorems 3.7 and 4.22, it holds

$$\frac{(\sigma_k^\dagger \text{diag } p^k \sigma_k, \tau_k^\dagger \text{diag } q^k \tau_k)}{\|(p^k, q^k)\|} \rightarrow (H^*, G^*).$$

Therefore, $(p^k, q^k)/\|(p^k, q^k)\|$ converges to $-(p^*, q^*)/\|(p^*, q^*)\|$. By Proposition 3.9 (3), we have

$$\frac{(p^k, q^k)}{k} = \frac{(p^k, q^k)}{\|(p^k, q^k)\|} \frac{\|(p^k, q^k)\|}{k} \rightarrow -\frac{1}{L} (p^*, q^*).$$

(2). By Theorems 3.7 and 4.22 and Proposition 3.9 (1), it holds

$$-\frac{1}{L} \left(\frac{1}{C_A} - \frac{1}{n} - \frac{1}{m} \right) = \lim_{k \rightarrow \infty} F_A(x_{k+1}, y_{k+1}) - F_A(x_k, y_k) = \lim_{k \rightarrow \infty} \frac{F_A(x_k, y_k)}{k}, \quad (4.34)$$

where the second equality follows from a well-known argument of Cesáro mean.

Since $e^{F_A(x_k, y_k)} = \text{tr} \sum_{\ell} x_k A_{\ell} y_k A_{\ell}^{\dagger} = \sum_{\ell, i, j} |(\sigma_k A_{\ell} \tau_k^{\dagger})_{ij}|^2 e^{p_i^k + q_j^k}$, we have

$$\sum_{\ell, i, j} |(\sigma_k A_{\ell} \tau_k^{\dagger})_{ij}|^2 e^{p_i^k + q_j^k - F_A(x_k, y_k)} = 1.$$

Suppose that (i, j) is in a lower-triangular block. By (1) and (4.34), $(p_i^k + q_j^k - F_A(x_k, y_k))/k$ converges to positive value $(1/L)(-p_i^* - q_j^* - 1/n - 1/m + 1/C) > 0$. Therefore, for some $c' > 0$ and $M' > 0$, it holds $p_i^k + q_j^k - F_A(x_k, y_k) \geq c'k$ for all $k > M'$. Then $|(\kappa_k A_{\ell} \sigma_k^{\dagger})_{ij}|^2 e^{c'k} \leq 1$ for all $k \geq M'$. $\square$

The same property (3) in Theorem 4.25 for $A^{(k)} := (e^{\text{diag } p^k/2} \sigma_k A \tau_k^{\dagger} e^{\text{diag } q^k/2}) / \|x_k^{1/2} A y_k^{1/2}\|$ is expected to hold by our numerical experiment but could not be proved due to the lack of the convergence guarantee of $[\pi(x_k^{1/2})v]$. We end this subsection with some implications of these results.

On finding a destabilizing 1-PSG. Suppose that A is not $(1/n, 1/m)$ -scalable. Consider $(X^*, Y^*) \in \mathcal{E}_A$ mapped to the extreme point (r^*, s^*) of $\text{Conv } \phi(\mathcal{S}_A)$ with the property that it maximizes r among all extreme points (r, s) maximizing $r + s$. The subspace pair (X^*, Y^*) violates (iii) in Theorem 4.16 and is a special certificate of unscalability, which is called *dominant* [19]. By Theorem 4.26 (2), after a large number k of iterations, the last r^* rows of σ_k and the first s^* rows of τ_k become bases of an ϵ -approximate dominant subspace $(X_{\epsilon}^*, Y_{\epsilon}^*)$ in the sense that $|u^{\top} A_{\ell} v| \leq \epsilon$ for all ℓ and all unit vectors $u \in X_{\epsilon}^*, v \in Y_{\epsilon}^*$. Franks, Soma, and Goemans [19] presented a procedure to round such an $e^{-p(n, m, N, b)}$ -approximate dominant subspace into the exact dominant subspace (X^*, Y^*) , where p is a polynomial and b is the bit-complexity of A . Hence, if we would establish *global* linear convergence in (2), a polynomial number of iterations of the gradient descent (4.20) would suffice to recover the dominant subspace and a destabilizing 1-PSG.

Matrix scaling case. An $n \times m$ matrix $M = (a_{ij})$ is viewed as a matrix tuple $A = (a_{ij} e_i e_j^{\top})_{ij: a_{ij} \neq 0}$. Consider the left-right action to A , in which the group is restricted to the subgroup $ST_n \times ST_m \subseteq SL_n \times SL_m$ consisting of diagonal matrices. The corresponding scaling problem is nothing but the matrix scaling problem of nonnegative matrix $(|a_{ij}|^2)$; see Section 3.3. The above results are also applicable to this setting. Indeed, the gradient ∇F_A is a pair of diagonal matrices. The trajectory of the gradient flow/descent is also diagonal in $P_n^1 \times P_m^1$, and equal to the one for the geometric programming objective (3.28) in matrix scaling. Then, all subspaces $X_{\alpha}, Y_{\alpha}, X_{\alpha, \beta}, Y_{\alpha, \beta}$ are coordinate subspaces. Hence a DM-decomposition B is obtained by row and column permutations, and is equivalent to the original (extended) DM-decomposition of M . In Theorem 4.26, unitary matrices σ_k, τ_k are permutation matrices, and all lower-triangular blocks become zero blocks after finitely many iterations. In particular, the expected convergence in Theorem 4.26 is true. This convergence property is almost the same as the one for Sinkhorn algorithm. Indeed, [28] showed that the limit (*Sinkhorn limit*) oscillates between the $(\mathbf{1}, \sum_{\alpha} (n_{\alpha}/m_{\alpha}) \mathbf{1}_{I_{\alpha}})$ -scaling B_r^* and $(\sum_{\alpha} (m_{\alpha}/n_{\alpha}) \mathbf{1}_{I_{\alpha}}, \mathbf{1})$ -scaling B_c^* of $\hat{B}$.

On the limit of operator Sinkhorn algorithm. This suggests an expectation of limiting behavior of the *operator Sinkhorn algorithm* (*Gurvits' algorithm*), the standard algorithm for the operator scaling problem. The operator Sinkhorn algorithm is viewed as alternating minimization of $F_A(x, y)$, where each step scales $A \rightarrow gA$ with $\tilde{\mu}(A) = (O, *)$

and $A \rightarrow Ah^\dagger$ with $\tilde{\mu}(A) = (*, O)$ alternatively. When it is applied to the $(p^* + \mathbf{1}/n, q^* + \mathbf{1}/m)$ -scaling B^* , the resulting scaling sequence oscillates between a $(\mathbf{1}, \sum_\alpha (n_\alpha/m_\alpha) \mathbf{1}_{I_\alpha})$ -scaling and $(\sum_\alpha (m_\alpha/n_\alpha) \mathbf{1}_{I_\alpha}, \mathbf{1})$ -scaling of B^* . With the view of Theorem 4.24 and the matrix scaling case above, it is reasonable to conjecture that it oscillates between orbits $U_n \cdot B_r^* \cdot U_m$ and $U_n \cdot B_c^* \cdot U_m$, where B_r^* (resp. B_c^*) is a $(\mathbf{1}, \sum_\alpha (n_\alpha/m_\alpha) \mathbf{1}_{I_\alpha})$ -scaling (resp. $(\sum_\alpha (m_\alpha/n_\alpha) \mathbf{1}_{I_\alpha}, \mathbf{1})$ -scaling) of $\hat{B}$.

4.3 Kronecker form of a matrix pencil

Finally we discuss a special case of $N = 2$, i.e., $A = (A_1, A_2)$. In this case, A is naturally identified with *matrix pencil* $sA_1 + A_2 \in \mathbb{C}(s)^{2(n \times m)}$, where s is an indeterminate. Here we reveal a connection to the *Kronecker canonical form* of $sA_1 + A_2$, and suggest a new numerical method for finding the Kronecker structure based on the gradient descent.

The pencil $sA_1 + A_2$ is called *regular* if $n = m$ and $\det(sA_1 + A_2) \neq 0$ for some $s \in \mathbb{C}$. Otherwise, the pencil $sA_1 + A_2$ is called *singular*. For simplicity, we assume (again) that $\ker A_1 \cap \ker A_2 = \{0\}$ and $\ker A_1^\dagger \cap \ker A_2^\dagger = \{0\}$. The Kronecker form is a canonical form of a (singular) pencil under transformations $g(sA_1 + A_2)h^\dagger$ for $g \in GL_n$ and $h \in GL_m$. The standard reference of the Kronecker form is [20, Chapter XII]; see also [41, Section 5.1.3] for its importance on system analysis. For a positive integer ϵ , define $\epsilon \times (\epsilon + 1)$ matrix L_ϵ by

$$(L_\epsilon)_{ij} := \begin{cases} 1 & \text{if } j = i, \\ s & \text{if } j = i + 1, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 4.27 (Kronecker form; [20, Chapter XII]). *There are $g \in GL_n, h \in GL_m$ such that*

$$g(sA_1 + A_2)h^\dagger = L_{\epsilon_1} \oplus L_{\epsilon_2} \oplus \cdots \oplus L_{\epsilon_c} \oplus (sC + D) \oplus L_{\eta_d}^\dagger \oplus L_{\eta_{d-1}}^\dagger \oplus \cdots \oplus L_{\eta_1}^\dagger, \quad (4.35)$$

where $sC + D$ is a regular pencil, and $\epsilon_1, \epsilon_2, \dots, \epsilon_c, \eta_1, \eta_2, \dots, \eta_d$ are positive integers determined as follows:

- ϵ_j is the minimum degree of a polynomial vector $x_j(s)$ in $\ker sA_1 + A_2$ that is linearly independent from $x_1(s), x_2(s), \dots, x_{j-1}(s)$ over $\mathbb{C}(s)$.
- η_j is the minimum degree of a polynomial vector $y_j(s)$ in $\ker(sA_1 + A_2)^\dagger$ that is linearly independent from $y_1(s), y_2(s), \dots, y_{j-1}(s)$ over $\mathbb{C}(s)$.

The indices $\epsilon_1 \leq \cdots \leq \epsilon_c, \eta_1 \leq \cdots \leq \eta_d$, called the *minimal indices*, are uniquely determined. If $n = m$ and $sA_1 + A_2$ is singular, then the Kronecker form has a zero block with sum of row and column sizes greater than n . Therefore, by Theorem 4.16, we have:

Corollary 4.28. *A pencil $sA_1 + A_2$ is regular if and only if $n = m$ and (A_1, A_2) is $(\mathbf{1}/n, \mathbf{1}/n)$ -scalable.*

We point out a further connection that the Kronecker form in the arrangement of (4.35) is viewed as almost a DM-decomposition in our sense. Let b denote the number of diagonal blocks of $gAh^\dagger$ in (4.35). For $\gamma \in [b]$, let I_γ and J_γ denote the row and column index sets, respectively, of the γ -th diagonal block of $gAh^\dagger$. Define X_γ by the vector subspace spanned by the rows of g of indices in $I_{\gamma+1} \cup I_{\gamma+2} \cup \cdots \cup I_b$. Similarly, define Y_γ by the vector subspace spanned by the rows of h having indices in $J_\gamma \cup \cdots \cup J_1$.

We let $(X_0, Y_0) := (\mathbb{C}^n, \{0\})$ (and $(X_b, Y_b) := (\{0\}, \mathbb{C}^m)$). Suppose that $sC + D (= g(sA_1 + A_2)h^\dagger[I_{c+1}, J_{c+1}])$ exists and is an $n_0 \times n_0$ upper-triangular matrix. Let Z_β denote the vector space spanned by the rows of g having the last $n_0 - \beta$ indices in I_{c+1} , and let W_β denote the vector space spanned by the rows of h having the first β indices in J_{c+1} . Let $X_{c,\beta} := X_{c+1} + Z_\beta$ and $Y_{c,\beta} := Y_c + W_\beta$, where $+$ is the direct sum. Consider all indices γ with $(|I_\gamma|, |J_\gamma|) \neq (|I_{\gamma+1}|, |J_{\gamma+1}|)$, and suppose that they are ordered as $0 =: \gamma_0 < \gamma_1 < \dots < \gamma_\theta := b$.

Proposition 4.29. (1) $\{(X_{\gamma_\alpha}, Y_{\gamma_\alpha})\}_{\alpha=0}^\theta$ is the coarse DM-flag of (A_1, A_2) .

(2) Suppose that $sC + D$ is an $n_0 \times n_0$ upper-triangular pencil. Then the union of $\{(X_\gamma, Y_\gamma)\}_{\gamma=0}^b$ and $\{(X_{c,\beta}, Y_{c,\beta})\}_{\beta=1}^{n_0-1}$ is a DM-flag of (A_1, A_2) .

Proof. (1). Suppose that $\mathcal{E}_A$ consists of (X'_α, Y'_α) for $\alpha = 0, 1, 2, \dots, \theta'$, arranged as in (4.22). We show $(X'_\alpha, Y'_\alpha) = (X_{\gamma_\alpha}, Y_{\gamma_\alpha})$ for $\alpha = 0, 1, 2, \dots, \theta' = \theta$. Consider the convex hull K_A of $(0, 0)$ and $\phi(X_\gamma, Y_\gamma)$ for all γ . Then K_A belongs to $\text{Conv } \phi(\mathcal{S}_A)$, and the maximal faces of K_A are composed of the line segments connecting points $\phi(X_\gamma, Y_\gamma)$ from $\gamma = 0$ to b with bending points $\phi(X_{\gamma_\alpha}, Y_{\gamma_\alpha})$.

We show $K_A = \text{Conv } \phi(\mathcal{S}_A)$ by the induction on the number b of diagonal blocks. Consider the base case $b = 1$ where the Kronecker form consists of a single block. It suffices to show $\mathcal{E}_A = \{(\mathbb{C}^n, 0), (0, \mathbb{C}^m)\}$. Suppose that $sA_1 + A_2$ is an $n_0 \times n_0$ regular pencil $sC + D$. By regularity, there is no $(X, Y) \in \mathcal{S}_A$ with $\dim X + \dim Y > n_0$ (otherwise $sA_1 + A_2$ is singular over $\mathbb{C}(s)$). This means no point in $\phi(\mathcal{S}_A)$ beyond the line segment between $(n_0, 0)$ and $(0, n_0)$. Therefore, we have $\mathcal{E}_A = \{(\mathbb{C}^n, 0), (0, \mathbb{C}^m)\}$. Suppose that $sA_1 + A_2 = L_n$. Suppose to the contrary that there is $(X, Y) \in \mathcal{S}_A$ with $\dim X/n + \dim Y/(n+1) > 1$. By basis change, we may assume that

$$sA_1 + A_2 = \begin{pmatrix} B & C \\ O & D \end{pmatrix},$$

where O is the $r \times s$ zero matrix for $(r, s) := (\dim X, \dim Y)$. By $r \geq 1$ and $r + s \geq n + 1$, B is a pencil of $n - r$ row and s columns with $s > n - r$. Then $\ker B$ contains a polynomial vector with degree at most $n - r < n$; use Cramer's formula to see this. Necessarily $\ker sA_1 + A_2$ also has such a polynomial vector. This is a contradiction to Theorem 4.27 ($\epsilon_1 = \epsilon_c = n$). The case $sA_1 + A_2 = L_n^\dagger$ is similar.

Consider a general case of $b \geq 2$. We can choose γ^*, α^* such that $0 < \gamma^* < b$, $0 < \alpha^* < \theta'$, and the line segment between $\phi(X_{\gamma^*}, Y_{\gamma^*})$ and $\phi(X'_{\alpha^*}, Y'_{\alpha^*})$ meets with K_A only at $\phi(X_{\gamma^*}, Y_{\gamma^*})$. Consider $(U, V) := (X_{\gamma^*} + X'_{\alpha^*}, Y_{\gamma^*} \cap Y'_{\alpha^*})$ and $(U', V') := (X_{\gamma^*} \cap X'_{\alpha^*}, Y_{\gamma^*} + Y'_{\alpha^*})$. By the construction and (4.21), one of $\phi(U, V)$ and $\phi(U', V')$ is outside of K_A . Suppose that $\phi(U, V) \notin K_A$. Consider submatrix $A' := (sA_1 + A_2)[\bigcup_{\gamma=1}^{\gamma^*} I_\gamma, \bigcup_{\gamma=1}^{\gamma^*} J_\gamma]$, that is also a Kronecker form with a smaller number of blocks. From $U \supseteq X_{\gamma^*}$, $V \subseteq Y_{\gamma^*}$, and $\phi(U, V) \notin K_A$, it necessarily holds $K_{A'} \neq \text{Conv } \phi(\mathcal{S}_{A'})$. However, this is a contradiction to the inductive assumption. The case $\phi(U', V') \notin K_A$ is similar; consider sub-Kronecker form $(sA_1 + A_2)[\bigcup_{\gamma=\gamma^*+1}^b I_\gamma, \bigcup_{\gamma=\gamma^*+1}^b J_\gamma]$.

(2). Observe that all integer points in the maximal faces of $\text{Conv } \phi(\mathcal{S}_A)$ are obtained by the images of (X_γ, Y_γ) and $(X_{c,\beta}, Y_{c,\beta})$. This implies that $\{(X_\gamma, Y_\gamma)\}_\gamma \cup \{(X_{c,\beta}, Y_{c,\beta})\}_\beta$ is a maximal chain of $\mathcal{L}_A$. $\square$

The matrix pencil $g(sA_1 + A_2)h^\dagger$ corresponding to a coarse DM-decomposition $g(A_1, A_2)h^\dagger$ is particularly called a *coarse Kronecker triangular form*. This is a refinement of a *quasi-Kronecker triangular form* in [7] and *generalized Schur form* in [16, 49] if g, h are unitary and $sC + D$ is triangular.

Then, the convergence (Theorem 4.26) of the gradient descent (4.20) can be applied as follows:

Theorem 4.30 (Convergence to a coarse Kronecker triangular form). *Let (x_k, y_k) be a solution of (4.20). Decompose $x_k = \sigma_k^\dagger e^{\text{diag } p^k} \sigma_k$ and $y_k = \tau_k^\dagger e^{\text{diag } q^k} \tau_k$, where σ_k and τ_k are unitary matrices, and p^k and q^k are nondecreasing and nonincreasing vectors, respectively. Then, $\sigma_k(sA_1 + A_2)\tau_k^\dagger$ accumulates to coarse Kronecker triangular forms, where the convergence is linear in the same sense as in Theorem 4.26 (3).*

A coarse Kronecker triangular form is enough for determining the structure of the Kronecker form. Indeed, each (non-square) rectangular diagonal block is a $k\nu \times k(\nu + 1)$ or $k(\nu + 1) \times k\nu$ matrix for some integers k, ν , from which all minimal indices $\epsilon_1, \epsilon_2, \dots, \epsilon_c, \eta_1, \eta_2, \dots, \eta_d$ can be identified.

The above theorem suggests an iterative method for determining the minimal indices of a singular pencil, which is based on the simple gradient descent and is conceptually different from the existing algorithms, e.g., [16, 49]. It is an interesting future direction to develop a numerically stable algorithm based on our results.

Acknowledgments

We thank Shin-ichi Ohta, Harold Nieuwboer, and Michael Walter for discussion, We also thank Taihei Oki and Tasuku Soma for comments and Shun Sato for suggesting [47]. The first author was supported by JSPS KAKENHI Grant Number JP21K19759, JP24K21315.

References

- [1] Z. Allen-Zhu, A. Garg, Y. Li, R. Oliveira, and A. Wigderson. Operator scaling via geodesically convex optimization, invariant theory and polynomial identity testing. In *Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing, STOC 2018*, pages 172–181. ACM, 2018.
- [2] F. Alvarez, J. Bolte, and O. Brahic. Hessian Riemannian gradient flows in convex programming. *SIAM J. Control Optim.*, 43(2):477–501, 2004.
- [3] N. Ambrosio, L. Gigli and G. Savaré. *Gradient Flows: In Metric Spaces and in the Space of Probability Measures*. Birkhäuser, Basel, 2 edition, 2008.
- [4] A. Auslender. How to deal with the unbounded in optimization: theory and algorithms. *Math. Programming*, 79(1-3):3–18, 1997.
- [5] M. Bačák. *Convex Analysis and Optimization in Hadamard Spaces*. De Gruyter, Berlin, 2014.
- [6] A. Beck. *First-Order Methods in Optimization*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2017.
- [7] T. Berger and S. Trenn. The quasi-Kronecker form for matrix pencils. *SIAM J. Matrix Anal. Appl.*, 33(2):336–368, 2012.

- [8] N. Boumal. *An Introduction to Optimization on Smooth Manifolds*. Cambridge University Press, Cambridge, 2023.
- [9] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer-Verlag, Berlin, 1999.
- [10] S. Bubeck. Convex optimization: Algorithms and complexity. *Found. Trends Mach. Learn.*, 8(3-4):231–357, 2015.
- [11] P. Bürgisser, C. Franks, A. Garg, R. Oliveira, M. Walter, and A. Wigderson. Efficient algorithms for tensor scaling, quantum marginals, and moment polytopes. In *59th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2018*, pages 883–897, 2018.
- [12] P. Bürgisser, C. Franks, A. Garg, R. Oliveira, M. Walter, and A. Wigderson. Towards a theory of non-commutative optimization: geodesic 1st and 2nd order methods for moment maps and polytopes. In *60th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2019*, pages 845–861, 2019.
- [13] P. Bürgisser, Y. Li, H. Nieuwboer, and M. Walter. Interior-point methods for unconstrained geometric programming and scaling problems. arXiv:2008.12110, 2020.
- [14] P.-E. Caprace and A. Lytchak. At infinity of finite-dimensional CAT(0) spaces. *Math. Ann.*, 346(1):1–21, 2010.
- [15] X. Chen and S. Sun. Calabi flow, geodesic rays, and uniqueness of constant scalar curvature Kähler metrics. *Ann. of Math. (2)*, 180(2):407–454, 2014.
- [16] J. Demmel and B. Kågström. The generalized Schur decomposition of an arbitrary pencil $A - \lambda B$: robust software with error bounds and applications. I. Theory and algorithms. *ACM Trans. Math. Software*, 19(2):160–174, 1993.
- [17] A. L. Dulmage and N. S. Mendelsohn. Coverings of bipartite graphs. *Canadian J. Math.*, 10:517–534, 1958.
- [18] C. Franks. Operator scaling with specified marginals. In *Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, pages 190–203. ACM, New York, 2018.
- [19] C. Franks, T. Soma, and M. X. Goemans. Shrunk subspaces via operator Sinkhorn iteration. In *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1655–1668. SIAM, Philadelphia, PA, 2023.
- [20] F. R. Gantmacher. *The Theory of Matrices. Vols. 1, 2*. Chelsea Publishing Co., New York, 1959.
- [21] A. Garg, L. Gurvits, R. Oliveira, and A. Wigderson. Algorithmic and optimization aspects of Brascamp-Lieb inequalities, via operator scaling. *Geom. Funct. Anal.*, 28(1):100–145, 2018.
- [22] A. Garg, L. Gurvits, R. Oliveira, and A. Wigderson. Operator scaling: theory and applications. *Found. Comput. Math.*, 20(2):223–290, 2020.

- [23] A. Garg and R. Oliveira. Recent progress on scaling algorithms and applications. *Bull. Eur. Assoc. Theor. Comput. Sci. EATCS*, (125):14–49, 2018.
- [24] V. Georgoulas, J. W. Robbin, and D. A. Salamon. *The Moment-Weight Inequality and the Hilbert-Mumford Criterion—GIT from the Differential Geometric Viewpoint*, volume 2297 of *Lecture Notes in Mathematics*. Springer, Cham, 2021.
- [25] V. Guillemin and S. Sternberg. Convexity properties of the moment mapping. II. *Invent. Math.*, 77(3):533–546, 1984.
- [26] L. Gurvits. Classical complexity and quantum entanglement. *J. Comput. System Sci.*, 69(3):448–484, 2004.
- [27] M. Hamada and H. Hirai. Computing the nc-rank via discrete convex optimization on CAT(0) spaces. *SIAM J. Appl. Algebra Geom.*, 5(3):455–478, 2021.
- [28] K. Hayashi, H. Hirai, and K. Sakabe. Finding Hall blockers by matrix scaling. *Mathematics of Operations Research*, 49:2166–2179, 2024.
- [29] H. Hirai. Convex analysis on Hadamard spaces and scaling problems. *Found. Comput. Math.*, 24:1979–2016, 2024.
- [30] H. Hirai, H. Nieuwboer, and M. Walter. Interior-point methods on manifolds: theory and applications. In *64th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2023*, pages 2021–2030. IEEE, 2023.
- [31] J.-B. Hiriart-Urruty and C. Lemaréchal. *Fundamentals of Convex Analysis*. Springer-Verlag, Berlin, 2001.
- [32] H. Ito, S. Iwata, and K. Murota. Block-triangularizations of partitioned matrices under similarity/equivalence transformations. *SIAM J. Matrix Anal. Appl.*, 15(4):1226–1255, 1994.
- [33] G. Ivanyos, Y. Qiao, and K. V. Subrahmanyam. Non-commutative Edmonds’ problem and matrix semi-invariants. *Comput. Complex.*, 26(3):717–763, 2017.
- [34] G. Ivanyos, Y. Qiao, and K. V. Subrahmanyam. Constructive non-commutative rank computation is in deterministic polynomial time. *Comput. Complex.*, 27(4):561–593, 2018.
- [35] Y. Iwamasa, T. Oki, and T. Soma. Algorithmic aspects of semistability of quiver representations. arXiv:2407.06493, 2024.
- [36] M. Kapovich, B. Leeb, and J. Millson. Convex functions on symmetric spaces, side lengths of polygons and the stability inequalities for weighted configurations at infinity. *J. Differential Geom.*, 81(2):297–354, 2009.
- [37] A. Karlsson and G. A. Margulis. A multiplicative ergodic theorem and nonpositively curved spaces. *Comm. Math. Phys.*, 208(1):107–123, 1999.
- [38] B. Kleiner and B. Leeb. Rigidity of invariant convex sets in symmetric spaces. *Invent. Math.*, 163(3):657–676, 2006.

- [39] T. C. Kwok, L. C. Lau, and A. Ramachandran. Spectral analysis of matrix scaling and operator scaling. *SIAM J. Comput.*, 50(3):1034–1102, 2021.
- [40] U. F. Mayer. Gradient flows on nonpositively curved metric spaces and harmonic maps. *Comm. Anal. Geom.*, 6(2):199–253, 1998.
- [41] K. Murota. *Matrices and Matroids for Systems Analysis*. Springer-Verlag, Berlin, 2000.
- [42] A. S. Nemirovsky and D. B. Yudin. *Problem Complexity and Method Efficiency in Optimization*. John Wiley & Sons, Inc., New York, 1983.
- [43] W. T. Obuchowska. On the minimizing trajectory of convex functions with unbounded level sets. *Comput. Optim. Appl.*, 27(1):37–52, 2004.
- [44] S. Ohta and M. Pálfi. Discrete-time gradient flows and law of large numbers in Alexandrov spaces. *Calc. Var. Partial Differential Equations*, 54(2):1591–1610, 2015.
- [45] R. T. Rockafellar. *Convex Analysis*. Princeton University Press, Princeton, NJ, 1970.
- [46] T. Sakai. *Riemannian Geometry*. American Mathematical Society, Providence, RI, 1996.
- [47] J. M. Sanz Serna and K. C. Zygalakis. Contractivity of Runge-Kutta methods for convex gradient systems. *SIAM J. Numer. Anal.*, 58(4):2079–2092, 2020.
- [48] R. Sinkhorn. A relationship between arbitrary positive matrices and doubly stochastic matrices. *Ann. Math. Statist.*, 35:876–879, 1964.
- [49] P. Van Dooren. The computation of Kronecker’s canonical form of a singular pencil. *Linear Algebra Appl.*, 27:103–140, 1979.
- [50] C. T. Woodward. Moment maps and geometric invariant theory. arXiv:0912.1132, 2011.

A Proof of Proposition 4.10

Identify $\mathfrak{g}$ with tangent space T_I of G at I , and define the right-invariant Riemannian metric $\langle, \rangle$ on G by $\langle X, Y \rangle := \operatorname{Re} \operatorname{tr} X g^{-1} (Y g^{-1})^\dagger$ for $X, Y \in T_g, g \in G$. Here we consider the following version of the Kempf-Ness function $\Phi_v : G \rightarrow \mathbb{R}$:

$$\Phi_v(g) := \frac{1}{2} \log \|\pi(g)v\|^2 \quad (g \in G). \quad (\text{A.1})$$

Lemma A.1. (1) $F_v(g^\dagger g) = 2\Phi_v(g)$.

(2) $\nabla \Phi_v(g) = \tilde{\mu}(\pi(g)v)g$; see [24, Theorem 4.1(i)].

(3) $2\nabla F_v(g^\dagger g) = g^\dagger \nabla \Phi_v(g) + \nabla \Phi_v(g)^\dagger g$.

Proof. (1) is obvious. (2). For $X \in T_g$, it holds

$$\begin{aligned}\langle \nabla \Phi_v(g), X \rangle &= \lim_{t \rightarrow 0} \frac{1}{2} \log \|\pi(e^{tXg^{-1}}g)v\|^2 = \operatorname{Re} \frac{(\pi(g)v, \Pi(Xg^{-1})\pi(g)v)}{(\pi(g)v, \pi(g)v)} \\ &= \operatorname{Re} \operatorname{tr} \tilde{\mu}(\pi(g)v)(Xg^{-1})^\dagger = \langle \tilde{\mu}(\pi(g)v)g, X \rangle.\end{aligned}$$

(3). For $H \in T_{g^\dagger g}(M)$, it holds

$$\begin{aligned}\langle \nabla F_v(g^\dagger g), H \rangle_{g^\dagger g} &= \lim_{t \rightarrow \infty} \frac{d}{dt} F_v(g^\dagger e^{tg^{-\dagger}Hg^{-1}}g) = \lim_{t \rightarrow \infty} \frac{d}{dt} 2\Phi_v(e^{tg^{-\dagger}Hg^{-1/2}}g) \\ &= \langle \nabla \Phi_v(g), g^{-\dagger}H \rangle_g = \langle g^\dagger \nabla \Phi_v(g) + \nabla \Phi_v(g)^\dagger g, H/2 \rangle_{g^\dagger g}.\end{aligned}$$

□

Lemma A.2 (see [24, Theorem 4.1(ii)]). *For $\zeta = [v] \in \mathbb{P}(V)$, define map $\varphi : G \rightarrow \mathbb{P}(V)$ by $g \mapsto g \cdot \zeta := [\pi(g)v]$. Then it holds*

$$d\varphi(g)\nabla \Phi_v(g) = \nabla \frac{\|\mu(g \cdot \zeta)\|^2}{2}.$$

Proof. The infinitesimal action of $X \in \mathfrak{g}$ at ζ is denoted by $X_\zeta^\# = (d/dt)e^{tX} \cdot \zeta|_{t=0} \in T_\zeta$. By definition of the moment map, for $\eta \in \mathbb{P}(V)$, $\xi \in T_\eta$, and $X \in \mathfrak{k}$, it holds

$$d\langle \mu(\eta), X \rangle \xi = \omega(X_\eta^\#, \xi) = \langle JX_\eta^\#, \xi \rangle,$$

where J is the complex structure and $\omega(\cdot, \cdot) = \langle J\cdot, \cdot \rangle$ is the symplectic form on $\mathbb{P}(V)$. By taking $X = \mu(\eta)$, it holds

$$d\frac{\|\mu(\eta)\|^2}{2}\xi = \langle J(\mu(\eta))_\eta^\#, \xi \rangle = \langle (i\mu(\eta))_\eta^\#, \xi \rangle.$$

In particular, $\nabla \|\mu(\eta)\|^2/2 = (i\mu(\eta))_\eta^\#$. On the other hand, it holds

$$d\varphi(g)X = \frac{d}{dt}(e^{tXg^{-1}}g) \cdot \zeta|_{t=0} = (Xg^{-1})_{g \cdot \zeta}^\#.$$

Letting $\eta = g \cdot \zeta = [\pi(g)v]$ and $X = \nabla \Phi_v(g) = \tilde{\mu}(\pi(g)v)g = i\mu(g \cdot \zeta)g$, we have the claim. □

Lemma A.3. *For $\zeta \in \mathbb{P}(V)$, define map $\tilde{\varphi} : M \rightarrow \mathbb{P}(V)$ by $x \mapsto x^{1/2} \cdot \zeta$. Then it holds*

$$d\tilde{\varphi}(x)\nabla F_v(x) = \nabla \frac{\|\mu(x^{1/2} \cdot \zeta)\|^2}{4}.$$

Proof. Define $\phi : M \rightarrow M$ by $x \mapsto x^{1/2}$. Then $\tilde{\varphi} = \varphi \circ \phi$, and hence $d\tilde{\varphi}(x)H = d\varphi(x^{1/2})(d\phi(x)H)$. On the other hand, ϕ^{-1} is given by $x \mapsto x^2$. From $d\phi^{-1} \circ d\phi = I$, we have

$$x^{1/2}(d\phi(x)H) + (d\phi(x)H)x^{1/2} = H \quad (\text{A.2})$$

Let $H := \nabla F_v(x)$. By Lemma A.1 (3), it must hold $\nabla F_v(x) = x^{1/2}\nabla \Phi_v(x^{1/2})/2 + \nabla \Phi_v(x^{1/2})^\dagger x^{1/2}/2$. By the unique solvability of Lyapunov equation (A.2), it must hold $d\phi(x)H = \nabla \Phi_v(x^{1/2})/2 (= \nabla \Phi_v(x^{1/2})^\dagger/2)$. Thus, we have $d\tilde{\varphi}(x)\nabla F_v(x) = d\varphi(x^{1/2})\nabla \Phi_v(x)/2$. By Lemma A.2, we have the claim. □

Thus, for the solution $x(t)$ of (4.6), $\zeta(t) := [\pi(x^{1/2}(2t))v]$ satisfies

$$\dot{\zeta}(t) = 2d\tilde{\varphi}(x(2t))\dot{x}(2t) = -2d\tilde{\varphi}(x(2t))\nabla F_v(x(2t)) = -\nabla \frac{\|\mu(\zeta(t))\|^2}{2}.$$